

Microsoft

(DP-600)

Implementing Analytics Solutions Using Microsoft Fabric (beta)

Total: **222 Questions**

Link: <https://certyiq.com/papers/microsoft/dp-600>

Question: 1

Case study -

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Overview -

Contoso, Ltd. is a US-based health supplements company. Contoso has two divisions named Sales and Research. The Sales division contains two departments named Online Sales and Retail Sales. The Research division assigns internally developed product lines to individual teams of researchers and analysts.

Existing Environment -

Identity Environment -

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Data Environment -

Contoso has the following data environment:

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The Research department uses an on-premises, third-party data warehousing product.

Fabric is enabled for contoso.com.

An Azure Data Lake Storage Gen2 storage account named storage1 contains Research division data for a product line named Productline1. The data is in the delta format.

A Data Lake Storage Gen2 storage account named storage2 contains Research division data for a product line named Productline2. The data is in the CSV format.

Requirements -

Planned Changes -

Contoso plans to make the following changes:

Enable support for Fabric in the Power BI Premium capacity used by the Sales division.

Make all the data for the Sales division and the Research division available in Fabric.

For the Research division, create two Fabric workspaces named Productline1ws and Productline2ws.

In Productline1ws, create a lakehouse named Lakehouse1.

In Lakehouse1, create a shortcut to storage1 named ResearchProduct.

Data Analytics Requirements -

Contoso identifies the following data analytics requirements:

All the workspaces for the Sales division and the Research division must support all Fabric experiences.

The Research division workspaces must use a dedicated, on-demand capacity that has per-minute billing.

The Research division workspaces must be grouped together logically to support OneLake data hub filtering based on the department name.

For the Research division workspaces, the members of ResearchReviewersGroup1 must be able to read lakehouse and warehouse data and shortcuts by using SQL endpoints.

For the Research division workspaces, the members of ResearchReviewersGroup2 must be able to read lakehouse data by using Lakehouse explorer.

All the semantic models and reports for the Research division must use version control that supports branching.

Data Preparation Requirements -

Contoso identifies the following data preparation requirements:

The Research division data for Productline1 must be retrieved from Lakehouse1 by using Fabric notebooks.

All the Research division data in the lakehouses must be presented as managed tables in Lakehouse explorer.

Semantic Model Requirements -

Contoso identifies the following requirements for implementing and managing semantic models:

The number of rows added to the Orders table during refreshes must be minimized.

The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -

Contoso identifies the following high-level requirements that must be considered for all solutions:

Follow the principle of least privilege when applicable.
Minimize implementation and maintenance effort when possible.
You need to ensure that Contoso can use version control to meet the data analytics requirements and the general requirements.
What should you do?

- A.Store at the semantic models and reports in Data Lake Gen2 storage.
- B.Modify the settings of the Research workspaces to use a GitHub repository.
- C.Modify the settings of the Research division workspaces to use an Azure Repos repository.
- D.Store all the semantic models and reports in Microsoft OneDrive.

Answer: C

Explanation:

C. Modify the settings of the Research division workspaces to use an Azure Repos repository.

Contoso needs version control for its data analytics and general requirements. In Microsoft Fabric and Power BI, version control is best managed through Git integration using Azure Repos or GitHub.

Azure Repos is a Git-based repository within Azure DevOps, providing versioning, tracking, and collaboration for code, reports, and semantic models.

This is the most appropriate choice for version-controlling Power BI artifacts like datasets, reports, and semantic models in a workspace.

Why the other options are incorrect:

Option A (Data Lake Gen2 storage): While Data Lake Gen2 stores big data files efficiently, it does not natively provide source/version control with branching mechanisms for workspace reports and semantic models. [1, 2, 3]

Option B (GitHub repository): Although Fabric supports GitHub, the case study specifically names the workspaces as the **Research division workspaces**. Option B incorrectly labels them as just "Research workspaces." Furthermore, inside enterprise Azure DevOps pipelines, Azure Repos is generally preferred for native Fabric lifecycle synchronization. [1, 4]

Option D (Microsoft OneDrive): OneDrive tracks historical changes to standalone files, but it does not support developer workflows like Git branching, merging, or code reviews needed for complex data analytics deployments. [1]

Reference:

<https://learn.microsoft.com/en-us/fabric/cicd/git-integration/intro-to-git-integration>

Question: 2

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HOTSPOT -

Case study -

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Existing Environment -

Identity Environment -

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Data Environment -

Contoso has the following data environment:

The Sales division uses a Microsoft Power BI Premium capacity.

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The Research department uses an on-premises, third-party data warehousing product.

Fabric is enabled for contoso.com.

An Azure Data Lake Storage Gen2 storage account named storage1 contains Research division data for a product line named Productline1. The data is in the delta format.

A Data Lake Storage Gen2 storage account named storage2 contains Research division data for a product line named Productline2. The data is in the CSV format.

Requirements -

Planned Changes -

Contoso plans to make the following changes:

Enable support for Fabric in the Power BI Premium capacity used by the Sales division.

Make all the data for the Sales division and the Research division available in Fabric.

For the Research division, create two Fabric workspaces named Productline1ws and Productline2ws.

In Productline1ws, create a lakehouse named Lakehouse1.

In Lakehouse1, create a shortcut to storage1 named ResearchProduct.

Data Analytics Requirements -

Contoso identifies the following data analytics requirements:

All the workspaces for the Sales division and the Research division must support all Fabric experiences.

The Research division workspaces must use a dedicated, on-demand capacity that has per-minute billing.

The Research division workspaces must be grouped together logically to support OneLake data hub filtering based on the department name.

For the Research division workspaces, the members of ResearchReviewersGroup1 must be able to read lakehouse and warehouse data and shortcuts by using SQL endpoints.

For the Research division workspaces, the members of ResearchReviewersGroup2 must be able to read lakehouse data by using Lakehouse explorer.

All the semantic models and reports for the Research division must use version control that supports branching.

Data Preparation Requirements -

Contoso identifies the following data preparation requirements:

The Research division data for Productline1 must be retrieved from Lakehouse1 by using Fabric notebooks.

All the Research division data in the lakehouses must be presented as managed tables in Lakehouse explorer.

Semantic Model Requirements -

Contoso identifies the following requirements for implementing and managing semantic models:

The number of rows added to the Orders table during refreshes must be minimized.

The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -

Contoso identifies the following high-level requirements that must be considered for all solutions:

Follow the principle of least privilege when applicable.

Minimize implementation and maintenance effort when possible.

You need to recommend a solution to group the Research division workspaces.

What should you include in the recommendation? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Grouping method:

	▼
Capacity	
Domain	
Tenant	

Tool:

	▼
OneLake data hub	
The Fabric Admin portal	
The Microsoft Entra admin center	

Answer:

Answer Area

Grouping method:

	▼
Capacity	
Domain	
Tenant	

Tool:

	▼
OneLake data hub	
The Fabric Admin portal	
The Microsoft Entra admin center	

Explanation:

Domain: Group the Research division workspaces based on their departmental context.

Domain allows you to group workspaces based on their purpose or business context. In this case, grouping by department (Research division) aligns with the requirement.

The Fabric Admin Portal is the primary management interface for Microsoft Fabric.

Why other answer are incorrect:

Capacity: While you can group things by capacity, "Capacity" usually refers to the compute and storage resources (the SKUs you purchase). Managing capacity is about resource allocation and performance, whereas "Domain" is specifically for the logical/business-oriented grouping and governance of workspaces.

Tenant: If the requirement is to view settings and metrics for the **entire organization** at the highest level, **Tenant** is used.

OneLake data hub: The data hub is used by end-users to view, filter, and discover data. While a user might see domain information there, they cannot use the data hub to create or administer the domains themselves.

Microsoft Entra admin center: This tool is for managing user identities, security groups, and permissions (e.g., Azure AD). While it handles identity access, it is not where you configure the specific logical grouping of Fabric workspaces into domains.

Reference:

<https://learn.microsoft.com/en-us/fabric/governance/domains#configure-domain-settings>

Question: 3

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Identity Environment -

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Requirements -

Planned Changes -

Contoso plans to make the following changes:

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Make all the data for the Sales division and the Research division available in Fabric.

For the Research division, create two Fabric workspaces named Productline1ws and Productline2ws.

In Productline1ws, create a lakehouse named Lakehouse1.

In Lakehouse1, create a shortcut to storage1 named ResearchProduct.

Data Analytics Requirements -

Contoso identifies the following data analytics requirements:

All the workspaces for the Sales division and the Research division must support all Fabric experiences.

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For the Research division workspaces, the members of ResearchReviewersGroup2 must be able to read lakehouse data by using Lakehouse explorer.
All the semantic models and reports for the Research division must use version control that supports branching.

Data Preparation Requirements -
Contoso identifies the following data preparation requirements:
The Research division data for Productline1 must be retrieved from Lakehouse1 by using Fabric notebooks.
All the Research division data in the lakehouses must be presented as managed tables in Lakehouse explorer.

Semantic Model Requirements -
Contoso identifies the following requirements for implementing and managing semantic models:
The number of rows added to the Orders table during refreshes must be minimized.
The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -
Contoso identifies the following high-level requirements that must be considered for all solutions:
Follow the principle of least privilege when applicable.
Minimize implementation and maintenance effort when possible.
You need to refresh the Orders table of the Online Sales department. The solution must meet the semantic model requirements.
What should you include in the solution?

- A. an Azure Data Factory pipeline that executes a Stored procedure activity to retrieve the maximum value of the OrderID column in the destination lakehouse
- B. an Azure Data Factory pipeline that executes a Stored procedure activity to retrieve the minimum value of the OrderID column in the destination lakehouse
- C. an Azure Data Factory pipeline that executes a dataflow to retrieve the minimum value of the OrderID column in the destination lakehouse
- D. an Azure Data Factory pipeline that executes a dataflow to retrieve the maximum value of the OrderID column in the destination lakehouse

Answer: D

Explanation:

D. an Azure Data Factory pipeline that executes a dataflow to retrieve the maximum value of the OrderID column in the destination lakehouse. A dataflow can be used to retrieve the max OrderID number (stored in the destination table - OrderID is a sequential number). This number can be used to set from which row data must be added to the destination table (implementing an incremental load).

Why the other options are incorrect:

Options B & C (Minimum value): Retrieving the minimum OrderID reflects the oldest historical order in the system. Relying on the minimum value would cause the refresh operation to continuously re-process and append existing rows, violating the requirement to minimize rows added during refreshes.

Option A (Stored procedure activity): Microsoft Fabric Lakehouses do not natively support executing standard T-SQL stored procedures to return scalar variables to a pipeline activity in the same direct manner as Dataflow Gen2 operations. A dataflow minimizes implementation and maintenance effort within the native Fabric interface.

Question: 4

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Requirements -

Planned Changes -

Contoso plans to make the following changes:

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Semantic Model Requirements -

Contoso identifies the following requirements for implementing and managing semantic models:

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The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -

Contoso identifies the following high-level requirements that must be considered for all solutions:

Follow the principle of least privilege when applicable.

Minimize implementation and maintenance effort when possible.

Which syntax should you use in a notebook to access the Research division data for Productline1?

- A.spark.read.format("delta").load("Tables/productline1/ResearchProduct")
- B.spark.sql("SELECT * FROM Lakehouse1.ResearchProduct ")
- C.external_table('Tables/ResearchProduct)
- D.external_table(ResearchProduct)

Answer: B

Explanation:

spark.sql("SELECT * FROM Lakehouse1.ResearchProduct ")

The syntax of C and D is correct for KQL databases (incorrect in this use-case). When the shortcut is created, no additional folders have been added to the Tables section, therefore answer A is incorrect. Once created, the line of answer B can be used to access data correctly.

Why the other options are incorrect

Option A: spark.read.format("delta").load("Tables/productline1/ResearchProduct") is an explicit physical path loading syntax. It relies on hardcoded string folders inside the storage structure rather than treating the dataset as a managed SQL table entity, increasing code maintenance overhead.

Options C & D: external_table() is not a standard native PySpark API or Spark SQL command used inside Microsoft Fabric notebooks for reading managed Lakehouse tables.

Reference:

<https://learn.microsoft.com/en-us/fabric/onelake/onelake-shortcuts>

Question: 5

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HOTSPOT -

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Overview -

Litware, Inc. is a manufacturing company that has offices throughout North America. The analytics team at Litware contains data engineers, analytics engineers, data analysts, and data scientists.

Existing Environment -

Fabric Environment -

Litware has been using a Microsoft Power BI tenant for three years. Litware has NOT enabled any Fabric capacities and features.

Available Data -

Litware has data that must be analyzed as shown in the following table.

Description	Original source	Total size
Customer data	Customer relationship management (CRM) system	50 MB
Product data	Customer relationship management (CRM) system	200 MB
Customer satisfaction surveys	SurveyMonkey	500 GB

The Product data contains a single table and the following columns.

Name	Data type
ProductID	Integer
ProductName	String
ProductCategory	String
ListPrice	Decimal

The customer satisfaction data contains the following tables:

Survey -

Question -

Response -

For each survey submitted, the following occurs:

One row is added to the Survey table.

One row is added to the Response table for each question in the survey.

The Question table contains the text of each survey question. The third question in each survey response is an overall satisfaction score. Customers can submit a survey after each purchase.

User Problems -

The analytics team has large volumes of data, some of which is semi-structured. The team wants to use Fabric to create a new data store.

Product data is often classified into three pricing groups: high, medium, and low. This logic is implemented in several databases and semantic models, but the logic does NOT always match across implementations.

Requirements -

Planned Changes -

Litware plans to enable Fabric features in the existing tenant. The analytics team will create a new data store as a proof of concept (PoC). The remaining Liware users will only get access to the Fabric features once the PoC is complete. The PoC will be completed by using a Fabric trial capacity

The following three workspaces will be created:

AnalyticsPOC: Will contain the data store, semantic models, reports pipelines, dataflow, and notebooks used to populate the data store

DataEngPOC: Will contain all the pipelines, dataflows, and notebooks used to populate OneLake

DataSciPOC: Will contain all the notebooks and reports created by the data scientists

The following will be created in the AnalyticsPOC workspace:

A data store (type to be decided)

A custom semantic model -

A default semantic model -

Interactive reports -

The data engineers will create data pipelines to load data to OneLake either hourly or daily depending on the data source. The analytics engineers will create processes to ingest, transform, and load the data to the data store in the AnalyticsPOC workspace daily. Whenever possible, the data engineers will use low-code tools for data ingestion. The choice of which data cleansing and transformation tools to use will be at the data engineers' discretion.

All the semantic models and reports in the Analytics POC workspace will use the data store as the sole data source.

Technical Requirements -

The data store must support the following:

Read access by using T-SQL or Python

Semi-structured and unstructured data

Row-level security (RLS) for users executing T-SQL queries

Files loaded by the data engineers to OneLake will be stored in the Parquet format and will meet Delta Lake specifications.

Data will be loaded without transformation in one area of the AnalyticsPOC data store. The data will then be cleansed, merged, and transformed into a dimensional model

The data load process must ensure that the raw and cleansed data is updated completely before populating the

dimensional model

The dimensional model must contain a date dimension. There is no existing data source for the date dimension. The Litware fiscal year matches the calendar year. The date dimension must always contain dates from 2010 through the end of the current year.

The product pricing group logic must be maintained by the analytics engineers in a single location. The pricing group data must be made available in the data store for T-SQL queries and in the default semantic model. The following logic must be used:

List prices that are less than or equal to 50 are in the low pricing group.

List prices that are greater than 50 and less than or equal to 1,000 are in the medium pricing group.

List prices that are greater than 1,000 are in the high pricing group.

Security Requirements -

Only Fabric administrators and the analytics team must be able to see the Fabric items created as part of the PoC.

Litware identifies the following security requirements for the Fabric items in the AnalyticsPOC workspace:

Fabric administrators will be the workspace administrators.

The data engineers must be able to read from and write to the data store. No access must be granted to datasets or reports.

The analytics engineers must be able to read from, write to, and create schemas in the data store. They also must be able to create and share semantic models with the data analysts and view and modify all reports in the workspace.

The data scientists must be able to read from the data store, but not write to it. They will access the data by using a Spark notebook

The data analysts must have read access to only the dimensional model objects in the data store. They also must have access to create Power BI reports by using the semantic models created by the analytics engineers.

The date dimension must be available to all users of the data store.

The principle of least privilege must be followed.

Both the default and custom semantic models must include only tables or views from the dimensional model in the data store. Litware already has the following Microsoft Entra security groups:

FabricAdmins: Fabric administrators

AnalyticsTeam: All the members of the analytics team

DataAnalysts: The data analysts on the analytics team

DataScientists: The data scientists on the analytics team

DataEngineers: The data engineers on the analytics team

AnalyticsEngineers: The analytics engineers on the analytics team

Report Requirements -

The data analysts must create a customer satisfaction report that meets the following requirements:

Enables a user to select a product to filter customer survey responses to only those who have purchased that product.

Displays the average overall satisfaction score of all the surveys submitted during the last 12 months up to a selected date.

Shows data as soon as the data is updated in the data store.

Ensures that the report and the semantic model only contain data from the current and previous year.

Ensures that the report respects any table-level security specified in the source data store.

Minimizes the execution time of report queries.

You need to assign permissions for the data store in the AnalyticsPOC workspace. The solution must meet the security requirements.

Which additional permissions should you assign when you share the data store? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

DataEngineers:

▼

Build Reports on the default dataset
Read All Apache Spark
Read All SQL analytics endpoint data

DataAnalysts:

▼

Build Reports on the default dataset
Read All Apache Spark
Read All SQL analytics endpoint data

DataScientists:

▼

Build Reports on the default dataset
Read All Apache Spark
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Answer:

Answer Area

DataEngineers: ▼

- Build Reports on the default dataset
- Read All Apache Spark
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- Build Reports on the default dataset
- Read All Apache Spark
- Read All SQL analytics endpoint data

DataScientists: ▼

- Build Reports on the default dataset
- Read All Apache Spark
- Read All SQL analytics endpoint data

Explanation:

Data Engineers: Read all Apache Spark - because they need to be able to work with Spark for Data curation.

Data Analysts: Build Reports on the default dataset - because they are report builders

Data Scientists: Read All SQL analytics Endpoints data_ They leverage curated data (by engineers) to do predictive analytics.

Incorrect answers:

Build Reports on the default dataset: Data engineers focus on backend data ingestion, transformation, and Lakehouse orchestration. Granting them front-end report-building privileges gives them unnecessary access to consumer layer components, violating the principle of least privilege.

Read All SQL analytics endpoint data: While SQL access allows read-only tabular querying, it does not grant the programmatic execution and file write privileges required to curate and manage underlying Delta tables using Spark notebooks.

Read All Apache Spark: Data analysts rely on a governed, structured semantic layer. Granting them programmatic read access to backend Spark storage exposes them to raw, unmodeled directory files, adding unnecessary data governance risk and bypassing table-level security.

Read All SQL analytics endpoint data: Analysts do not need direct connection pathways to the underlying SQL database engine since their reporting workloads are entirely fulfilled by building visuals on top of the auto-generated default semantic model.

Build Reports on the default dataset: Data scientists build predictive and statistical algorithms rather than

design client-facing monitoring dashboards, making reporting permissions unnecessary.

Read All Apache Spark: Data scientists in this scenario are meant to consume data after it has been cleansed and curated by the engineering team. Allowing them to read the raw Spark system gives them overly permissive access to operational folders before the data is formally structured.

Question: 6

CertyIQ

HOTSPOT -

Case study -

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The analytics team has large volumes of data, some of which is semi-structured. The team wants to use Fabric to create a new data store.

Product data is often classified into three pricing groups: high, medium, and low. This logic is implemented in several databases and semantic models, but the logic does NOT always match across implementations.

Requirements -

Planned Changes -

Litware plans to enable Fabric features in the existing tenant. The analytics team will create a new data store as a proof of concept (PoC). The remaining Liware users will only get access to the Fabric features once the PoC is complete. The PoC will be completed by using a Fabric trial capacity

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Semi-structured and unstructured data

Row-level security (RLS) for users executing T-SQL queries

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The data load process must ensure that the raw and cleansed data is updated completely before populating the dimensional model

The dimensional model must contain a date dimension. There is no existing data source for the date dimension. The Litware fiscal year matches the calendar year. The date dimension must always contain dates from 2010 through the end of the current year.

The product pricing group logic must be maintained by the analytics engineers in a single location. The pricing group data must be made available in the data store for T-SOL. queries and in the default semantic model. The following logic must be used:

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Enables a user to select a product to filter customer survey responses to only those who have purchased that product.

Displays the average overall satisfaction score of all the surveys submitted during the last 12 months up to a selected dat.

Shows data as soon as the data is updated in the data store.

Ensures that the report and the semantic model only contain data from the current and previous year.

Ensures that the report respects any table-level security specified in the source data store.

Minimizes the execution time of report queries.

You need to create a DAX measure to calculate the average overall satisfaction score.

How should you complete the DAX code? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
Rolling 12 Overall Satisfaction =
```

```
VAR NumberOfMonths = 12
```

```
VAR LastCurrentDate = MAX ( 'Date' [Date] )
```

```
VAR Period = DATESINPERIOD ( 'Date' [Date], LastCurrentDate,  
- NumberOfMonths, MONTH )
```

```
VAR Result =
```

```
    CALCULATE (
```

AVERAGE('Survey'[Response Value]),
AVERAGEA('Question'[Question Text]),
AVERAGEX(VALUES('Survey'[Customer Key]),

LastCurrentDate,
NumberOfMonths,
Period,

```
    'Survey Question' [Question Title] = "Overall Satisfaction"  
)
```

```
RETURN
```

```
    Result
```

Answer:

Answer Area

```
Rolling 12 Overall Satisfaction =  
VAR NumberOfMonths = 12  
VAR LastCurrentDate = MAX ( 'Date' [Date] )  
VAR Period = DATESINPERIOD ( 'Date' [Date], LastCurrentDate,  
- NumberOfMonths, MONTH )  
VAR Result =  
    CALCULATE (  
        AVERAGE('Survey'[Response Value]),  
        AVERAGEA('Question'[Question Text]),  
        AVERAGEX(VALUES('Survey'[Customer Key]),  
        LastCurrentDate,  
        NumberOfMonths,  
        Period,  
        'Survey Question'[Question Title] = "Overall Satisfaction"  
    )  
    RETURN  
    Result
```

Explanation:

1. Average(Survey[Response Value]) as this one is the only value that can be aggregated.

Customer Key Does not make any sense , as it just a key to identify a customer, this can never provide us the score.

2. Period : Variable is defined as to select 1 year of date range. Can directly be passed in the Filter context of Calculate formula.

Incorrect answers:

AVERAGEA('Question'[Question Text]) Reason: Question Text contains descriptive strings (e.g., "How satisfied were you with your checkout experience?"). You cannot mathematically average alphanumeric text strings to find a metric calculation.

AVERAGEX(VALUES('Survey'[Customer Key]), ...) Reason: Customer Key is an ID column used to distinguish individual users, not the actual satisfaction rating. Iterating over distinct customer keys instead of summarizing response records directly would change the statistical grain and yield wrong calculation outcomes.

LastCurrentDate This variable holds a singular scalar value (e.g., just the single maximum date like 2026-06-24). If passed into CALCULATE, it would restrict the formula to look at exactly one day instead of a continuous 12-month window.

NumberOfMonths This variable holds a static scalar integer (12). CALCULATE requires valid column filtering expressions or tables; passing a single constant integer will result in a syntax evaluation error.

Question: 7

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Ensures that the report respects any table-level security specified in the source data store.

Minimizes the execution time of report queries.

You need to resolve the issue with the pricing group classification.

How should you complete the T-SQL statement? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

CREATE [dbo].[ProductsWithPricingGroup]

FUNCTION
PROCEDURE
TABLE
VIEW

AS

```
SELECT ProductId,  
       ProductName,  
       ProductCategory,  
       ListPrice,
```

CASE
COALESCE
IIF
SET

```
    WHEN ListPrice <= 50 THEN 'low'
```

WHEN(ListPrice > 50 AND ListPrice < 1000) THEN 'medium'
WHEN(ListPrice >= 50 AND ListPrice < 1000) THEN 'medium'
WHEN ListPrice BETWEEN 50 AND 1000) THEN 'medium'

```
    WHEN ListPrice > 1000 THEN 'high'
```

```
END AS PricingGroup  
FROM dbo.Products
```

Answer:

```

CREATE [dropdown] [dbo].[ProductsWithPricingGroup]
    FUNCTION
    PROCEDURE
    TABLE
    VIEW
AS

SELECT ProductId,
        ProductName,
        ProductCategory,
        ListPrice,
        [dropdown]
        CASE
        COALESCE
        IIF
        SET
        WHEN ListPrice <= 50 THEN 'low'
        [dropdown]
        WHEN(ListPrice > 50 AND ListPrice < 1000) THEN 'medium'
        WHEN(ListPrice >= 50 AND ListPrice < 1000) THEN 'medium'
        WHEN ListPrice BETWEEN 50 AND 1000) THEN 'medium'
        WHEN ListPrice > 1000 THEN 'high'
    END AS PricingGroup
FROM dbo.Products

```

Explanation:

VIEW: from an existing table.

* CASE: correct syntax before the WHENs.

* WHEN ListPrice BETWEEN 50 AND 1000 THEN 'medium': the other two options miss value 1000; on the other hand, the BETWEEN includes both 50 and 1000.

Incorrect answers:

Why the TABLE is incorrect because a standard CREATE TABLE structure requires explicit column schema type definitions inside parentheses, or it must use the CREATE TABLE ... AS SELECT (CTAS) pattern which does not utilize an AS keyword prior to the SELECT statement. PROCEDURE and FUNCTION do not match this specific definition sequence.

COALESCE simply checks an array for the first non-null argument, IIF handles simple ternary true/false pairs without WHEN syntax, and SET is used to assign values to variables rather than inside a select list projection.

Both alternative choices end with < 1000. If a product costs exactly 1000, those choices would evaluate to false, leaving a gap where a price of 1000 would slip through unclassified to the next check.

Question: 8

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What should you recommend using to ingest the customer data into the data store in the AnalyticsPOC workspace?

A.a stored procedure

B.a pipeline that contains a KQL activity

C.a Spark notebook

D.a dataflow

Answer: D

Explanation:

D. a dataflow.

Even though the text reads "Data will be loaded without transformation in one area of the AnalyticsPOC data store": in general, dataflows are used when data transformations are involved after ingestion. As suggested by user BHARAT, the Copy Activity should be the optimal solution.

Why the Other Options Are Incorrect:

A. Stored procedure

Requires database-level control

Higher maintenance

Not Fabric-native ingestion method

B. Pipeline with KQL activity

Used for orchestration or KQL transformations

Overkill for structured customer ingestion

More engineering effort

C. Spark notebook

Requires code

Higher maintenance

Not minimal implementation effort

Question: 9

CertyIQ

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Litware plans to enable Fabric features in the existing tenant. The analytics team will create a new data store as a proof of concept (PoC). The remaining Liware users will only get access to the Fabric features once the PoC is complete. The PoC will be completed by using a Fabric trial capacity

The following three workspaces will be created:

AnalyticsPOC: Will contain the data store, semantic models, reports pipelines, dataflow, and notebooks used to populate the data store

DataEngPOC: Will contain all the pipelines, dataflows, and notebooks used to populate OneLake

DataSciPOC: Will contain all the notebooks and reports created by the data scientists

The following will be created in the AnalyticsPOC workspace:

A data store (type to be decided)

A custom semantic model -

A default semantic model -

Interactive reports -

The data engineers will create data pipelines to load data to OneLake either hourly or daily depending on the data source. The analytics engineers will create processes to ingest, transform, and load the data to the data store in the AnalyticsPOC workspace daily. Whenever possible, the data engineers will use low-code tools for data ingestion. The choice of which data cleansing and transformation tools to use will be at the data engineers' discretion.

All the semantic models and reports in the Analytics POC workspace will use the data store as the sole data source.

Technical Requirements -

The data store must support the following:

Read access by using T-SQL or Python

Semi-structured and unstructured data

Row-level security (RLS) for users executing T-SQL queries

Files loaded by the data engineers to OneLake will be stored in the Parquet format and will meet Delta Lake specifications.

Data will be loaded without transformation in one area of the AnalyticsPOC data store. The data will then be cleansed, merged, and transformed into a dimensional model

The data load process must ensure that the raw and cleansed data is updated completely before populating the dimensional model

The dimensional model must contain a date dimension. There is no existing data source for the date dimension. The Litware fiscal year matches the calendar year. The date dimension must always contain dates from 2010 through the end of the current year.

The product pricing group logic must be maintained by the analytics engineers in a single location. The pricing group data must be made available in the data store for T-SQL queries and in the default semantic model. The following logic must be used:

List prices that are less than or equal to 50 are in the low pricing group.

List prices that are greater than 50 and less than or equal to 1,000 are in the medium pricing group.

List prices that are greater than 1,000 are in the high pricing group.

Security Requirements -

Only Fabric administrators and the analytics team must be able to see the Fabric items created as part of the PoC. Litware identifies the following security requirements for the Fabric items in the AnalyticsPOC workspace:

Fabric administrators will be the workspace administrators.

The data engineers must be able to read from and write to the data store. No access must be granted to datasets or reports.

The analytics engineers must be able to read from, write to, and create schemas in the data store. They also must be able to create and share semantic models with the data analysts and view and modify all reports in the workspace.

The data scientists must be able to read from the data store, but not write to it. They will access the data by using a Spark notebook

The data analysts must have read access to only the dimensional model objects in the data store. They also must have access to create Power BI reports by using the semantic models created by the analytics engineers.

The date dimension must be available to all users of the data store.

The principle of least privilege must be followed.

Both the default and custom semantic models must include only tables or views from the dimensional model in the data store. Litware already has the following Microsoft Entra security groups:

FabricAdmins: Fabric administrators

AnalyticsTeam: All the members of the analytics team

DataAnalysts: The data analysts on the analytics team

DataScientists: The data scientists on the analytics team

DataEngineers: The data engineers on the analytics team

AnalyticsEngineers: The analytics engineers on the analytics team

Report Requirements -

The data analysts must create a customer satisfaction report that meets the following requirements:

Enables a user to select a product to filter customer survey responses to only those who have purchased that product.

Displays the average overall satisfaction score of all the surveys submitted during the last 12 months up to a selected date.

Shows data as soon as the data is updated in the data store.

Ensures that the report and the semantic model only contain data from the current and previous year.

Ensures that the report respects any table-level security specified in the source data store.

Minimizes the execution time of report queries.

Which type of data store should you recommend in the AnalyticsPOC workspace?

- A.a data lake
- B.a warehouse
- C.a lakehouse
- D.an external Hive metastore

Answer: C

Explanation:

A lakehouse The data store must handle semi-structured and unstructured data, therefore a Lakehouse should be the optimal solution supporting read access with T-SQL and Python.

Why the other options are incorrect

Option A (a data lake): A data lake stores raw files (such as raw text or blobs), but lacks the structured relational engine layer, SQL endpoint interfaces, and optimization tools needed to natively enforce table-level security or minimize reporting tool execution times.

Option B (a warehouse): A data warehouse is highly optimized for structured data and relational models, but it does not natively support unstructured files, failing the case study's file-handling conditions.

Option D (an external Hive metastore): An external Hive metastore is a metadata registry used to organize database architecture parameters for Spark/Hive ecosystems. It is not an actual unified data storage layer.

Question: 10

CertyIQ

You have a Fabric warehouse that contains a table named Staging.Sales. Staging.Sales contains the following columns.

Name	Data type	Nullable
ProductID	Integer	No
ProductName	Varchar(30)	No
SalesDate	Datetime2(6)	No
WholesalePrice	Decimal(18,2)	Yes
Amount	Decimal(18,2)	Yes

You need to write a T-SQL query that will return data for the year 2023 that displays ProductID and ProductName and has a summarized Amount that is higher than 10,000.

Which query should you use?

A.

```
SELECT ProductID, ProductName, SUM(Amount) AS TotalAmount
FROM Staging.Sales
WHERE DATEPART(YEAR,SaleDate) = '2023'
GROUP BY ProductID, ProductName
HAVING SUM(Amount) > 10000
```

B.

```
SELECT ProductID, ProductName, SUM(Amount) AS TotalAmount
FROM Staging.Sales
GROUP BY ProductID, ProductName
HAVING DATEPART(YEAR,SaleDate) = '2023' AND SUM(Amount) > 10000
```

- C.
- ```
SELECT ProductID, ProductName, SUM(Amount) AS TotalAmount
FROM Staging.Sales
WHERE DATEPART(YEAR,SaleDate) = '2023' AND SUM(Amount) > 10000
```
- D.
- ```
SELECT ProductID, ProductName, SUM(Amount) AS TotalAmount
FROM Staging.Sales
WHERE DATEPART(YEAR,SaleDate) = '2023'
GROUP BY ProductID, ProductName
HAVING TotalAmount > 10000
```

Answer: A

Explanation:

```
SELECT ProductID, ProductName, SUM(Amount) AS TotalAmount
FROM Staging.Sales
WHERE DATEPART(YEAR, SaleDate) = '2023'
GROUP BY ProductID, ProductName
HAVING SUM(Amount) > 10000
```

Answer A is the only answer with valid syntax.

Answer B is excluded because the filter on year should be included in the WHERE-clause. It is the result of a function, but it is not an aggregate.

Answer C is excluded because the filter on SUM(Amount) should be included in the HAVING-clause, since this is an aggregate.

Answer D is excluded because the HAVING-clause does not accept aliases.

Question: 11

CertyIQ

HOTSPOT -

You have a data warehouse that contains a table named Stage.Customers. Stage.Customers contains all the customer record updates from a customer relationship management (CRM) system. There can be multiple updates per customer.

You need to write a T-SQL query that will return the customer ID, name, postal code, and the last updated time of the most recent row for each customer ID.

How should you complete the code? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
WITH CUSTOMERBASE AS (  
    SELECT [CustomerID]  
    , [CustomerName]  
    , [PostalCode]  
    , [LastUpdated]  
    , x =  OVER (PARTITION BY CustomerID  
                                ORDER BY LastUpdated DESC)  
    FROM [LakehousePOC].[dbo].[CustomerChanges]  
    )  
    SELECT CustomerID, CustomerName, PostalCode, LastUpdated  
    FROM CUSTOMERBASE  
  
  
Having Max(LastUpdated) = 1  
WHERE LastUpdated = Max(LastUpdated)  
WHERE X = 1
```

Answer:

Answer Area

```
WITH CUSTOMERBASE AS (  
    SELECT [CustomerID]  
    , [CustomerName]  
    , [PostalCode]  
    , [LastUpdated]  
    , x =  OVER (PARTITION BY CustomerID  
                                ORDER BY LastUpdated DESC)  
    FROM [LakehousePOC].[dbo].[CustomerChanges]  
    )  
    SELECT CustomerID, CustomerName, PostalCode, LastUpdated  
    FROM CUSTOMERBASE  
  
  
Having Max(LastUpdated) = 1  
WHERE LastUpdated = Max(LastUpdated)  
WHERE X = 1
```

Explanation:

Box 1: **ROW_NUMBER()**

ROW_NUMBER() is a window function that assigns a unique sequential number to each row within a partition, based on the ORDER BY clause.

In this case, it is used to assign a row number to each CustomerID, ordering by LastUpdated DESC.

This ensures that the most recent record for each CustomerID gets ROW_NUMBER() = 1.

Box 2: **WHERE X = 1**

The alias X stores the row number assigned by ROW_NUMBER().

Since we only want the most recent row per CustomerID, we filter for X = 1.

This approach ensures we get only the latest entry for each customer, based on LastUpdated.

Question: 12

CertyIQ

HOTSPOT -

You have a Fabric tenant.

You plan to create a Fabric notebook that will use Spark DataFrames to generate Microsoft Power BI visuals.

You run the following code.

```
from powerbiclient import QuickVisualize, get_dataset_config, Report
```

```
PBI_visualize = QuickVisualize(get_dataset_config(df))
```

```
PBI_visualize
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
The code embeds an existing Power BI report.	☐	☐
The code creates a Power BI report.	☐	☐
The code displays a summary of the DataFrame.	☐	☐

Answer:

Answer Area

Statements	Yes	No
The code embeds an existing Power BI report.	☐	☒
The code creates a Power BI report.	☒	☐
The code displays a summary of the DataFrame.	☐	☒

Explanation:

"The code embeds an existing Power BI report."

Answer: **No**

Embedding a Power BI report typically involves using the Power BI Embedded API, which is not indicated here.

"The code creates a Power BI report."

Answer: **Yes**

If the code generates a new Power BI report (e.g., using Python, Power BI REST API, or Power BI Desktop automation), then this is correct.

"The code displays a summary of the DataFrame."

Answer: **No**

Displaying a summary of a DataFrame would involve functions like `df.describe()`, `df.info()`, or `df.head()` in Pandas. Since this is about Power BI, this function is not relevant.

Question: 13

CertyIQ

You are the administrator of a Fabric workspace that contains a lakehouse named Lakehouse1. Lakehouse1 contains the following tables:

Table1: A Delta table created by using a shortcut

Table2: An external table created by using Spark

Table3: A managed table -

You plan to connect to Lakehouse1 by using its SQL endpoint.

What will you be able to do after connecting to Lakehouse1?

- A. Read Table3.
- B. Update the data Table3.
- C. Read Table2.
- D. Update the data in Table1.

Answer: A

Explanation:

A. Read Table3.

When you connect to a Lakehouse SQL endpoint in Microsoft Fabric, you are connecting to:

The SQL analytics endpoint (read-only T-SQL surface over lakehouse tables)

The SQL endpoint has important limitations.

Why the other options are incorrect

Option B & D (Update data): As stated above, the SQL analytics endpoint functions in a read-only state. Modifying, altering, or updating row entries requires switching back to a Spark notebook or utilizing Dataflow Gen2 destination pipelines.

Option C (Read Table2): External tables defined via raw Apache Spark schema configurations are not automatically synchronized or exposed to the SQL analytics endpoint metadata layer. To make external table formats accessible via T-SQL queries, they must be registered inside the workspace using explicit OneLake table shortcuts.

Question: 14

CertyIQ

You have a Fabric tenant that contains a warehouse.

You use a dataflow to load a new dataset from OneLake to the warehouse.

You need to add a PowerQuery step to identify the maximum values for the numeric columns.

Which function should you include in the step?

- A. Table.MaxN
- B. Table.Max
- C. Table.Range
- D. Table.Profile

Answer: D

Explanation:

The Table.Profile function in PowerQuery is specifically designed to provide statistical information about a table's columns, including the maximum values for numeric columns. When you apply Table.Profile to your dataset, it generates a summary of each column, which includes the minimum, maximum, average, standard deviation, count of distinct values, and other useful statistics.

Why the other options are incorrect

Option A (Table.MaxN): This function filters the source to return the top **N number of rows** based on specific condition criteria, rather than outputting maximum statistical measurements for multiple numeric columns.

Option B (Table.Max): This function scans the table to return **one single row** containing the highest value found in one specific column criteria. It cannot generate a multi-column summary of maximum statistical values across all numeric columns simultaneously.

Option C (Table.Range): This function retrieves a subset or offset window block of rows starting from a

specified index row number. It does not perform mathematical maximum or statistical calculations on data elements.

Reference:

<https://learn.microsoft.com/en-us/powerquery-m/table-profile>

Question: 15

CertyIQ

You have a Fabric tenant that contains a machine learning model registered in a Fabric workspace. You need to use the model to generate predictions by using the PREDICT function in a Fabric notebook. Which two languages can you use to perform model scoring? Each correct answer presents a complete solution. NOTE: Each correct answer is worth one point.

- A.T-SQL
- B.DAX
- C.Spark SQL
- D.PySpark

Answer: CD

Explanation:

Spark SQL: This language leverages the Spark framework, which is commonly used for big data processing and machine learning. Spark SQL allows you to run SQL queries over data and includes support for machine learning model scoring with the PREDICT function.

PySpark: This is the Python API for Spark. PySpark allows you to interact with Spark using Python, which is one of the most popular programming languages for data analysis and machine learning. PySpark includes capabilities for integrating with machine learning models and running predictions using the PREDICT function.

Why Not the Other Options:

T-SQL: This is Transact-SQL, primarily used for managing and querying data in SQL Server and Azure SQL Database. It doesn't support the PREDICT function for machine learning model scoring in Fabric notebooks.

DAX: This is Data Analysis Expressions, a language used in Power BI, Power Pivot, and Analysis Services for data modeling and calculations. It doesn't support the PREDICT function for machine learning model scoring in Fabric notebooks.

Question: 16

CertyIQ

You are analyzing the data in a Fabric notebook. You have a Spark DataFrame assigned to a variable named df. You need to use the Chart view in the notebook to explore the data manually. Which function should you run to make the data available in the Chart view?

- A.displayHTML
- B.show
- C.write
- D.display

Answer: D

Explanation:

Display: This function is specifically designed for rendering visual representations of data within interactive notebooks. When you use the display function on a Spark DataFrame, it enables the data to be shown in a tabular format and also makes it available for interactive charting and visualization within the notebook.

Why the other options are incorrect:

Option A (displayHTML): This function is used to render raw HTML strings, custom JavaScript, or standalone iframe components inside the notebook output cell. It does not interface with the notebook's built-in interactive dataframe chart engine.

Option B (show): The standard PySpark command `df.show()` prints out a raw text-based representation of the data directly into the cell logs. It cannot be parsed by the visual interface and does not provide an interactive Chart view toggle.

Option C (write): The `.write` method is used for data persistence workloads (e.g., saving a DataFrame out into long-term storage as Delta, Parquet, or CSV files). It does not present data within the workspace user interface.

Question: 17

CertyIQ

You have a Fabric tenant that contains a Microsoft Power BI report named Report1. Report1 includes a Python visual.

Data displayed by the visual is grouped automatically and duplicate rows are NOT displayed.

You need all rows to appear in the visual.

What should you do?

- A. Reference the columns in the Python code by index.
- B. Modify the Sort Column By property for all columns.
- C. Add a unique field to each row.
- D. Modify the Summarize By property for all columns.

Answer: C

Explanation:

C. Add a unique field to each row.

Add a unique field to each row: When you add a unique identifier (such as an ID column or a combination of columns that uniquely identifies each row), Power BI recognizes that each row is unique and will display all rows in the visual without grouping them.

Why the other options are incorrect:

Option A (Reference columns by index): Python script editors reference the automatically provided dataframe columns by string labels or standard index positions, but this only changes how you parse the data after it arrives in Python. It does not stop the Power BI service layer from dropping duplicates beforehand.

Option B (Modify the Sort Column By property): Changing sorting priorities alters the visual order of items on

screen but does not modify the underlying grain or record footprint of the imported dataframe.

Option D (Modify the Summarize By property): While setting column configurations to "Don't Summarize" changes individual column aggregation definitions, if multiple identical rows exist across all fields, Power BI will still consolidate them into a single unique dataframe row representation.

Question: 18

CertyIQ

DRAG DROP -

You have a Fabric tenant that contains a semantic model. The model contains data about retail stores.

You need to write a DAX query that will be executed by using the XMLA endpoint. The query must return a table of stores that have opened since December 1, 2023.

How should you complete the DAX expression? To answer, drag the appropriate values to the correct targets.

Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Values

Answer Area

DEFINE

EVALUATE

FILTER

SUMMARIZE

TABLE

```
VAR_SalesSince =  
    DATE (2023, 12, 01)
```

```
FILTER (
```

```
(Store, Store[Name], Store[OpenDate]),  
    Store[OpenDate] >= _SalesSince  
)
```

Answer:

Values

Answer Area

DEFINE

EVALUATE

FILTER

SUMMARIZE

TABLE

DEFINE

```
VAR_SalesSince =  
    DATE (2023, 12, 01)
```

EVALUATE

```
FILTER (
```

SUMMARIZE

```
(Store, Store[Name], Store[OpenDate]),  
    Store[OpenDate] >= _SalesSince  
)
```

Explanation:

DEFINE

This section defines a variable (VAR _SalesSince) that stores a specific date (2023-12-01).

EVALUATE

This keyword is required for executing and returning a table result in DAX queries.

SUMMARIZE

Used to create a summary table with specific columns.

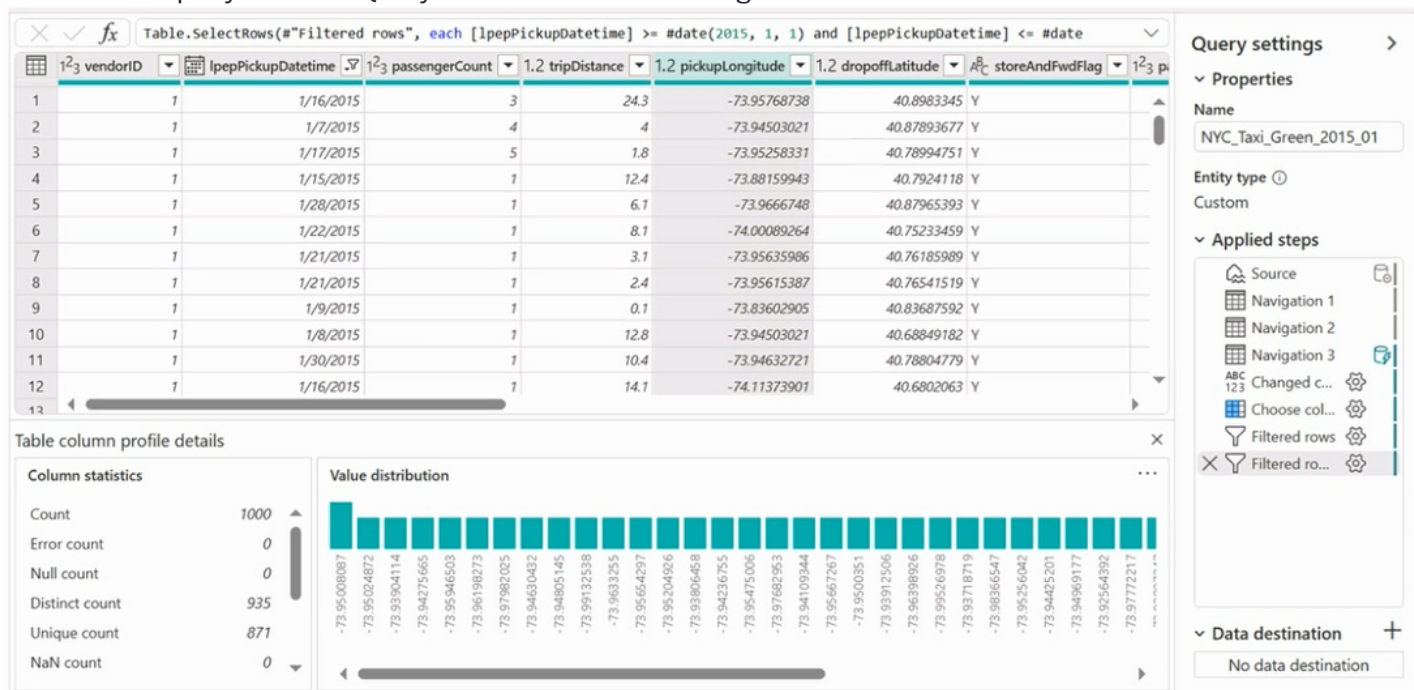
Here, it summarizes data from the Store table, including Store[Name] and Store[OpenDate].

Question: 19

CertyIQ

You have a Fabric workspace named Workspace1 that contains a dataflow named Dataflow1. Dataflow1 has a query that returns 2,000 rows.

You view the query in Power Query as shown in the following exhibit.



What can you identify about the pickupLongitude column?

- A. The column has duplicate values.
- B. All the table rows are profiled.
- C. The column has missing values.
- D. There are 935 values that occur only once.

Answer: A

Explanation:

Distinct Value: This refers to all different values present in a dataset. When you retrieve distinct values from a column, you eliminate duplicate values so that each value is shown once. For example, if a column contains the values [1, 2, 2, 3, 3, 3], the distinct values would be [1, 2, 3].

Unique Value: This usually refers to values that appear only once in the dataset. Unlike distinct values, a unique value will only be considered if it has no duplicates at all. For example, if a column contains the values [1, 2, 2, 3, 3, 3], the unique values would be [1], since only 1 appears without repetition.

Why the Other Options are Incorrect

B. All the table rows are profiled: Incorrect. The query states that the table returns 2,000 rows, but Power Query profiles only the top 1,000 rows by default.

C. The column has missing values: Incorrect. The column profile indicators show 0% empty or error values.

D. There are 935 values that occur only once: Incorrect. The number 935 represents the Distinct count (total number of different values), not the Unique count (values that appear exactly one time).

Question: 20

CertyIQ

You have a Fabric tenant named Tenant1 that contains a workspace named WS1. WS1 uses a capacity named C1 and contains a dataset named DS1.

You need to ensure read-write access to DS1 is available by using XMLA endpoint.

What should be modified first?

- A.the DS1 settings
- B.the WS1 settings
- C.the C1 settings
- D.the Tenant1 settings

Answer: C

Explanation:

C. the C1 settings.

To enable read-write access via the XMLA endpoint, the XMLA endpoint must be set to Read/Write at the capacity level.

In Power BI / Fabric:

XMLA endpoint configuration is controlled at the Premium capacity level

Workspaces inherit this setting from the capacity.

Why the Other Options Are Incorrect:

A. DS1 settings.

Dataset settings do not control XMLA endpoint mode

B. WS1 settings.

Workspace does not override capacity-level XMLA configuration

D. Tenant1 settings.

Tenant settings enable XMLA feature availability globally

But read/write mode is controlled at the capacity level

Question: 21

CertyIQ

You have a Fabric tenant that contains a workspace named Workspace1. Workspace1 is assigned to a Fabric

capacity.

You need to recommend a solution to provide users with the ability to create and publish custom Direct Lake semantic models by using external tools. The solution must follow the principle of least privilege.

Which three actions in the Fabric Admin portal should you include in the recommendation? Each correct answer presents part of the solution.

NOTE: Each correct answer is worth one point.

A.From the Tenant settings, set Allow XMLA Endpoints and Analyze in Excel with on-premises datasets to Enabled.

B.From the Tenant settings, set Allow Azure Active Directory guest users to access Microsoft Fabric to Enabled.

C.From the Tenant settings, select Users can edit data model in the Power BI service.

D.From the Capacity settings, set XMLA Endpoint to Read Write.

E.From the Tenant settings, set Users can create Fabric items to Enabled.

F.From the Tenant settings, enable Publish to Web.

Answer: ADE

Explanation:

A Allow XMLA Endpoints and Analyze in Excel with on-premises datasets.

From the Tenant settings, set Allow XMLA Endpoints and Analyze in Excel with on-premises datasets to Enabled. This allows users to interact with the dataset via XMLA endpoints and analyze data using Excel.

D. Set XMLA Endpoint to Read Write.

In the Capacity settings, configure the XMLA Endpoint to Read Write. This ensures that users have the necessary permissions to create and modify semantic models through external tools.

E Enable Users to Create Fabric Items.

From the Tenant settings, set Users can create Fabric items to Enabled. This grants users the ability to create custom semantic models within the Fabric workspace.

Question: 22

CertyIQ

You are creating a semantic model in Microsoft Power BI Desktop.

You plan to make bulk changes to the model by using the Tabular Model Definition Language (TMDL) extension for Microsoft Visual Studio Code.

You need to save the semantic model to a file.

Which file format should you use?

A.PBIP

B.PBIX

C.PBIT

D.PBIDS

Answer: A

Explanation:

A. PBIP.

PBIP (Power BI Project) is a file format that supports the open-source TMDL format and is designed for integrating Power BI with external development environments like Visual Studio Code.

PBIX is the common Power BI report file format but is not intended for bulk edits through TMDL.

PBIT is a Power BI template file, used for creating new reports based on an existing structure but not for bulk editing in Visual Studio Code.

PBIDS is for creating Power BI dataset connections and is unrelated to TMDL editing.

Question: 23

CertyIQ

HOTSPOT -

You have a Fabric tenant that contains a warehouse named Warehouse1. Warehouse1 contains three schemas named schemaA, schemaB, and schemaC.

You need to ensure that a user named User1 can truncate tables in schemaA only.

How should you complete the T-SQL statement? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

GRANT

	▼
ALTER CONNECT EXECUTE	

 ON

	▼
DATABASE::schemaA OBJECT::schemaA SCHEMA::schemaA	

 TO User1;

Answer:

Answer Area

GRANT

	▼
ALTER CONNECT EXECUTE	

 ON

	▼
DATABASE::schemaA OBJECT::schemaA SCHEMA::schemaA	

 TO User1;

Explanation:

ALTER: Allows modification of schema objects .

SCHEMA::schemaA: Grants permissions on an entire schema instead of an individual database or object .

Question: 24

CertyIQ

You plan to deploy Microsoft Power BI items by using Fabric deployment pipelines. You have a deployment pipeline that contains three stages named Development, Test, and Production. A workspace is assigned to each stage.

You need to provide Power BI developers with access to the pipeline. The solution must meet the following

requirements:

Ensure that the developers can deploy items to the workspaces for Development and Test.

Prevent the developers from deploying items to the workspace for Production.

Follow the principle of least privilege.

Which three levels of access should you assign to the developers? Each correct answer presents part of the solution.

NOTE: Each correct answer is worth one point.

- A. Build permission to the production semantic models
- B. Admin access to the deployment pipeline
- C. Viewer access to the Development and Test workspaces
- D. Viewer access to the Production workspace
- E. Contributor access to the Development and Test workspaces
- F. Contributor access to the Production workspace

Answer: BDE

Explanation:

Admin access to the deployment pipeline : Admin access to the deployment pipeline is necessary to manage the pipeline stages and deployment processes. This allows developers to configure deployments between stages but does not grant direct access to the workspaces themselves.

Viewer access to the Production workspace : Providing Viewer access to the Production workspace ensures that developers can see the items in the Production stage but cannot make changes or deploy items to the Production workspace. This restriction aligns with the requirement to prevent developers from deploying to the Production workspace.

Contributor access to the Development and Test workspaces : Contributor access to the Development and Test workspaces allows developers to deploy and modify items within these stages. This access is necessary to enable developers to work on and test their items before they move to production.

Why Other Options Are Not Selected:

A. Build permission to the production semantic models: This does not relate directly to the deployment pipeline access requirements and does not provide the needed permissions for deployment.

C. Viewer access to the Development and Test workspaces: Viewer access would only allow developers to see the items without making any changes or deployments, which does not meet the requirement to deploy items to these workspaces.

F. Contributor access to the Production workspace: This would grant developers the ability to deploy and modify items in the Production workspace, which contradicts the requirement to prevent developers from deploying to the Production stage.

Question: 25

CertyIQ

You have a Fabric workspace that contains a DirectQuery semantic model. The model queries a data source that has 500 million rows.

You have a Microsoft Power BI report named Report1 that uses the model. Report1 contains visuals on multiple pages.

You need to reduce the query execution time for the visuals on all the pages.

What are two features that you can use? Each correct answer presents a complete solution,

NOTE: Each correct answer is worth one point.

- A.user-defined aggregations
- B.automatic aggregation
- C.query caching
- D.OneLake integration

Answer: AB

Explanation:

A. User-defined aggregations.

Aggregations help improve query performance by pre-aggregating data at different levels (e.g., summarizing sales by month instead of querying raw transactions).

Users define aggregations based on expected queries, reducing the need to scan massive datasets.

B. Automatic aggregations.

Automatic aggregations optimize query performance by learning from query patterns and storing frequently used aggregate results in memory.

This reduces the load on the underlying data source, speeding up DirectQuery performance.

Question: 26

CertyIQ

You have a Fabric tenant that contains 30 CSV files in OneLake. The files are updated daily. You create a Microsoft Power BI semantic model named Model1 that uses the CSV files as a data source. You configure incremental refresh for Model1 and publish the model to a Premium capacity in the Fabric tenant. When you initiate a refresh of Model1, the refresh fails after running out of resources. What is a possible cause of the failure?

- A.Query folding is occurring.
- B.Only refresh complete days is selected.
- C.XMLA Endpoint is set to Read Only.
- D.Query folding is NOT occurring.
- E.The delta type of the column used to partition the data has changed.

Answer: D

Explanation:

D. Query folding is NOT occurring.

Incremental refresh depends on query folding to work efficiently.

When query folding works:

Power BI pushes the RangeStart and RangeEnd filters down to the data source.

Only the required partitions are processed.

Memory usage is controlled.

However, with CSV files in OneLake:

Query folding typically does NOT occur.

The entire dataset may be loaded into memory first.

Filtering happens after loading.

This can cause excessive memory consumption.

Result: Refresh fails due to running out of resources, even on Premium capacity.

Question: 27

CertyIQ

You have a Fabric tenant that uses a Microsoft Power BI Premium capacity.
You need to enable scale-out for a semantic model.
What should you do first?

- A. At the semantic model level, set Large dataset storage format to Off.
- B. At the tenant level, set Create and use Metrics to Enabled.
- C. At the semantic model level, set Large dataset storage format to On.
- D. At the tenant level, set Data Activator to Enabled.

Answer: C

Explanation:

C. At the semantic model level, set Large dataset storage format to On.

To enable scale-out for a semantic model in Power BI Premium capacity, the model must first:

Use Large dataset storage format

Scale-out allows:

Multiple read-only replicas of the semantic model

Improved query performance

Better concurrency handling.

Question: 28

CertyIQ

You have a Fabric tenant that contains a warehouse. The warehouse uses row-level security (RLS).
You create a Direct Lake semantic model that uses the Delta tables and RLS of the warehouse.
When users interact with a report built from the model, which mode will be used by the DAX queries?

- A. DirectQuery
- B. Dual
- C. Direct Lake
- D. Import

Answer: A

Explanation:

A. Direct Query "Row-level security only applies to queries on a Warehouse or SQL analytics endpoint in Fabric. Power BI queries on a warehouse in Direct Lake mode will fall back to Direct Query mode to abide by row-level security."

When users interact with a report built from a Direct Lake semantic model, which leverages row-level security (RLS) and Delta tables from a warehouse, the DAX queries will operate in DirectQuery mode.

Reference:

<https://learn.microsoft.com/en-us/fabric/data-warehouse/row-level-security>

Question: 29

CertyIQ

You have a Fabric tenant that contains a complex semantic model. The model is based on a star schema and contains many tables, including a fact table named Sales.

You need to create a diagram of the model. The diagram must contain only the Sales table and related tables. What should you use from Microsoft Power BI Desktop?

- A. data categories
- B. Data view
- C. Model view
- D. DAX query view

Answer: C

Explanation:

C. Model view.

Model view in Microsoft Power BI Desktop allows you to visualize and manage relationships between tables in a semantic model.

Since the requirement is to create a diagram that includes only the Sales table and its related tables, Model view is the best choice.

In Model view, you can filter and arrange tables to focus only on relevant parts of the data model, making it easier to analyze relationships.

Why not the other options?

A. Data categories → Used to define how data is treated (e.g., geography, URLs) but not for diagramming models.

B. Data view → Used for viewing raw table data but does not provide a relationship diagram.

D. DAX query view → Used to write and test DAX queries, but not for visualizing model relationships.

Question: 30

CertyIQ

You have a Fabric tenant that contains a semantic model. The model uses Direct Lake mode. You suspect that some DAX queries load unnecessary columns into memory. You need to identify the frequently used columns that are loaded into memory. What are two ways to achieve the goal? Each correct answer presents a complete solution. NOTE: Each correct answer is worth one point.

- A. Use the Analyze in Excel feature.
- B. Use the Vertipaq Analyzer tool.
- C. Query the `$$System.DISCOVER_STORAGE_TABLE_COLUMN_SEGMENTS` dynamic management view (DMV).
- D. Query the `DISCOVER_MEMORYGRANT` dynamic management view (DMV).

Answer: BC

Explanation:

B. Use the Vertipaq Analyzer tool.

Vertipaq Analyzer: This tool helps analyze the internal structure of your Power BI model. It provides detailed information about the storage and memory usage of your model, including which columns are frequently accessed and loaded into memory. This can help you identify unnecessary columns that are consuming resources.

Steps:

Export your Power BI model to a .pbix file.

Open the .pbix file in Power BI Desktop.

Use the Vertipaq Analyzer tool to analyze the model and review the column usage statistics.

C. Query the `$$System.DISCOVER_STORAGE_TABLE_COLUMN_SEGMENTS` dynamic management view (DMV).

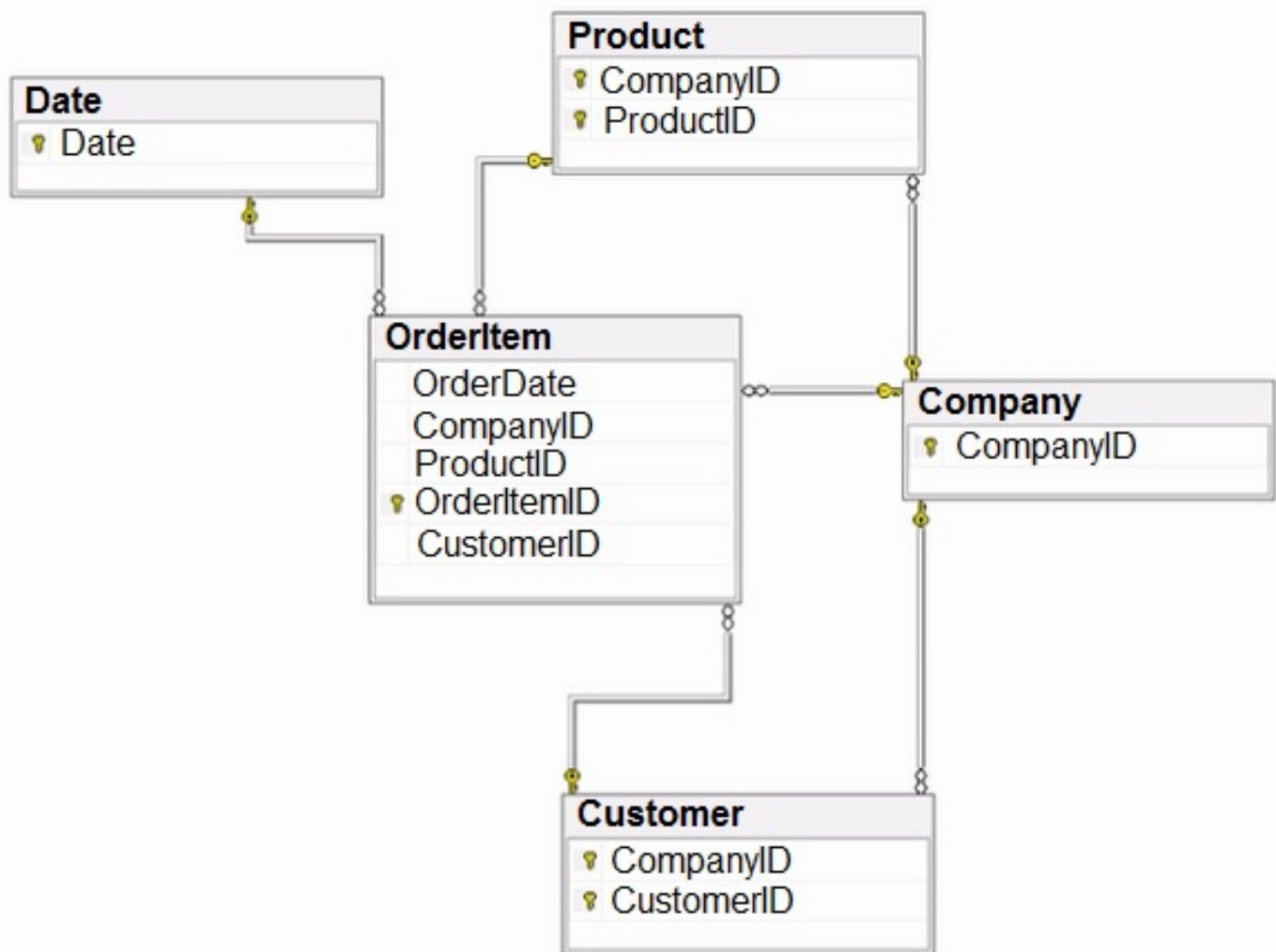
DMVs: Dynamic Management Views (DMVs) provide detailed information about the operations of your Power BI models. Specifically, the `$$System.DISCOVER_STORAGE_TABLE_COLUMN_SEGMENTS` DMV can give you insights into the storage and usage patterns of individual columns within your model.

Question: 31

CertyIQ

HOTSPOT -

You have the source data model shown in the following exhibit.



The primary keys of the tables are indicated by a key symbol beside the columns involved in each key. You need to create a dimensional data model that will enable the analysis of order items by date, product, and customer.

What should you include in the solution? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

The relationship between OrderItem and Product must be based on:

- ☐ The ProductID column
- ☐ Both the CompanyID and the ProductID columns
- ☐ A new key that combines the CompanyID and ProductID columns

The Company entity must be:

- ☐ Omitted
- ☐ Denormalized into the Product entity only
- ☐ Denormalized into the Customer and Product entities

Answer:

Answer Area

The relationship between OrderItem and Product must be based on:

The ProductID column
Both the CompanyID and the ProductID columns
A new key that combines the CompanyID and ProductID columns

The Company entity must be:

Omitted
Denormalized into the Product entity only
Denormalized into the Customer and Product entities

Explanation:

The Product ID column.

In a proper star schema:

OrderItem is a fact table.

Product is a dimension table.

The relationship should use the primary key of the dimension.

Denormalized into the Customer and Product entities.

In star schema modeling.

Snowflake schemas (separate Company table linked to Product and Customer) are not ideal in Power BI.

Power BI performs best with denormalized dimensions.

Question: 32

CertyIQ

You have a Fabric tenant that contains a semantic model named Model1. Model1 uses Import mode. Model1 contains a table named Orders. Orders has 100 million rows and the following fields.

Name	Data type	Description
OrderID	Integer	Column imported from the source
OrderDateTime	Date/Time	Column imported from the source
Quantity	Integer	Column imported from the source
Price	Decimal	Column imported from the source
TotalSalesAmount	Decimal	Calculated column that multiplies Quantity and Price
TotalQuantity	Integer	Measure

You need to reduce the memory used by Model1 and the time it takes to refresh the model. Which two actions should you perform? Each correct answer presents part of the solution.

NOTE: Each correct answer is worth one point.

- A.Split OrderDateTime into separate date and time columns.
- B.Replace TotalQuantity with a calculated column.
- C.Convert Quantity into the Text data type.

D.Replace TotalSalesAmount with a measure.

Answer: AD

Explanation:

A. Split OrderDateTime into separate date and time columns .

In Import mode, Power BI compresses data more efficiently when Date and Time are stored separately instead of a single DateTime column.

DateTime values have higher cardinality (more unique values), which increases memory usage.

Splitting into Date and Time reduces the number of unique values, improving compression and reducing memory usage.

D. Replace TotalSalesAmount with a measure .

Measures in Power BI are calculated at query time and do not consume additional memory, unlike calculated columns, which take up storage.

TotalSalesAmount is likely derived from Quantity * UnitPrice, so instead of storing it as a column, it should be computed dynamically using a DAX measure.

This reduces model size and improves performance.

Question: 33

CertyIQ

You have a Fabric tenant that contains a semantic model.

You need to prevent report creators from populating visuals by using implicit measures.

What are two tools that you can use to achieve the goal? Each correct answer presents a complete solution.

NOTE: Each correct answer is worth one point.

A.Microsoft Power BI Desktop

B.Tabular Editor

C.Microsoft SQL Server Management Studio (SSMS)

D.DAX Studio

Answer: AB

Explanation:

A. Microsoft Power BI Desktop: Power BI Desktop allows you to control and manage how measures are used within your reports. By carefully defining and using explicit measures within your data model, you can ensure that report creators use only these predefined measures instead of creating implicit measures automatically.

B. Tabular Editor: Tabular Editor is a powerful tool for managing and editing Power BI and Analysis Services tabular models. It allows you to enforce best practices, such as disabling implicit measures, by modifying the model's properties and ensuring that only explicit measures are available for use in reports.

Question: 34

CertyIQ

HOTSPOT -

You have a Fabric tenant that contains two lakehouses.

You are building a dataflow that will combine data from the lakehouses. The applied steps from one of the queries in the dataflow is shown in the following exhibit.

Query settings >

▼ Properties

Name

Customers1

Entity type ⓘ

Custom

▼ **Applied steps**

- Source
- Navigation 1
- Capitalized each word
- Appended query
- Changed column type
- Added custom
- Filtered rows**

Context menu for 'Filtered rows':

- Edit settings
- Rename
- Delete
- Delete until end
- Insert step after
- Move before
- Move after
- Extract previous...
- View data source query
- View query plan
- Properties

Use the drop-down menus to select the answer choice that completes each statement based on the information presented in the graphic.

NOTE: Each correct selection is worth one point.

Answer Area

[Answer choice] of the transformation steps in the query will fold.

▼

All
None
Some

The Added custom step will be performed in [answer choice].

▼

each lakehouse's query engine
the Microsoft Power Query engine
the source lakehouse query engine

Answer:

Answer Area

[Answer choice] of the transformation steps in the query will fold.

▼

All
None
Some

The Added custom step will be performed in [answer choice].

▼

each lakehouse's query engine
the Microsoft Power Query engine
the source lakehouse query engine

Explanation:

1. **Some** of the transformation steps in the query will fold."

Query folding refers to the ability of Power Query to push transformations back to the data source rather than processing them in Power Query itself.

The answer "Some".

Some transformations are foldable (executed at the source).

Some transformations are not foldable and will be executed locally in Power Query.

This is common because basic transformations (filtering, joins, aggregations, etc.) often fold, but more complex operations (custom columns, advanced functions, etc.) may not.

2. The Added custom step will be performed in **the Microsoft Power Query engine**.

When a transformation cannot be folded, it is processed in the Power Query engine instead of being pushed to the source.

Since the question mentions an added custom step, it likely includes:

Custom calculations

Complex transformations (like M code or certain joins)

Operations that cannot be executed at the data source

Such transformations are executed locally within Power Query rather than at the source (lakehouse query

engine).

Question: 35

You have a Fabric tenant that contains a lakehouse named Lakehouse1. Lakehouse1 contains a table named Table1. You are creating a new data pipeline.

You plan to copy external data to Table1. The schema of the external data changes regularly.

You need the copy operation to meet the following requirements:

Replace Table1 with the schema of the external data.

Replace all the data in Table1 with the rows in the external data.

You add a Copy data activity to the pipeline.

What should you do for the Copy data activity?

- A. From the Source tab, add additional columns.
- B. From the Destination tab, set Table action to Overwrite.
- C. From the Settings tab, select Enable staging.
- D. From the Source tab, select Enable partition discovery.
- E. From the Source tab, select Recursively.

Answer: B

Explanation:

B. From the Destination tab, set Table action to Overwrite.

You need the copy operation to:

Replace Table1 with the schema of the external data

Replace all data in Table1 with the external data

The best way to achieve this is by setting Table action to "Overwrite" in the Destination tab of the Copy Data activity.

"Overwrite" ensures that the table is completely replaced, including the schema and data, which meets both requirements.

This means that every time the pipeline runs, Table1 will have the latest schema and data from the external source.

Why not the other options?

A. From the Source tab, add additional columns

Adding columns does not replace the schema; it only appends extra fields.

This does not ensure that the table structure matches the external data.

C. From the Settings tab, select Enable staging

Staging is useful for performance improvements in large data transfers, but it does not affect schema replacement.

D. From the Source tab, select Enable partition discovery

Partition discovery helps when reading partitioned data, but it does not affect replacing the table schema or

data.

E. From the Source tab, select Recursively

Recursion is used for reading multiple files in hierarchical folders but does not replace the schema or data in the destination table.

Question: 36

CertyIQ

You have a Fabric tenant that contains a lakehouse.

You plan to query sales data files by using the SQL endpoint. The files will be in an Amazon Simple Storage Service (Amazon S3) storage bucket.

You need to recommend which file format to use and where to create a shortcut.

Which two actions should you include in the recommendation? Each correct answer presents part of the solution.

NOTE: Each correct answer is worth one point.

- A. Create a shortcut in the Files section.
- B. Use the Parquet format
- C. Use the CSV format.
- D. Create a shortcut in the Tables section.
- E. Use the delta format.

Answer: DE

Explanation:

D. Create Shortcut in Tables Section.

The SQL endpoint:

Can only query objects inside the Tables area of a lakehouse

Cannot directly query items in the Files section

Therefore:

The S3 shortcut must point to Delta files

The shortcut must be created under Tables

This registers the data as a table accessible via SQL.

E. Delta Format .

The SQL endpoint in Fabric only supports querying tables, not raw files.

Delta format:

Is the native table format in Fabric

Is fully supported by the SQL endpoint

Supports ACID transactions

Supports schema enforcement

Enables better performance.

Question: 37

You have a Fabric tenant that contains a lakehouse named Lakehouse1. Lakehouse1 contains a subfolder named Subfolder1 that contains CSV files.

You need to convert the CSV files into the delta format that has V-Order optimization enabled.

What should you do from Lakehouse explorer?

- A. Use the Load to Tables feature.
- B. Create a new shortcut in the Files section.
- C. Create a new shortcut in the Tables section.
- D. Use the Optimize feature.

Answer: A

Explanation:

A. Use the "Load to Tables" feature.

To convert CSV files into Delta format with V-Order optimization, you need to load the files into tables in the Lakehouse.

"Load to Tables" is a feature in Microsoft Fabric's Lakehouse Explorer that allows you to convert files (such as CSV) into managed Delta tables.

During this process, the data is converted into the Delta format, which supports V-Order optimization for better performance.

V-Order is an advanced columnar storage format that enhances query performance and compression for Delta tables.

Reference:

<https://learn.microsoft.com/en-us/fabric/data-engineering/load-to-tables#load-to-table-capabilities-overview>

Question: 38

You have a Fabric tenant that contains a lakehouse named Lakehouse1. Lakehouse1 contains an unpartitioned table named Table1.

You plan to copy data to Table1 and partition the table based on a date column in the source data.

You create a Copy activity to copy the data to Table1.

You need to specify the partition column in the Destination settings of the Copy activity.

What should you do first?

- A. From the Destination tab, set Mode to Append.
- B. From the Destination tab, select the partition column.
- C. From the Source tab, select Enable partition discovery.
- D. From the Destination tabs, set Mode to Overwrite.

Answer: D

Explanation:**D. From the Destination tabs, set Mode to Overwrite.**

Expand Advanced, in Table action, select Overwrite, and then select Enable partition, under Partition columns, select Add column, and choose the column you want to use as the partition column. You can choose to use a single column or multiple columns as the partition column.

Reference:

<https://learn.microsoft.com/en-us/fabric/data-factory/tutorial-lakehouse-partition#load-data-to-lakehouse-using-partition-columns>

Question: 39**CertyIQ**

HOTSPOT -

You have a Fabric tenant that contains a warehouse named Warehouse1. Warehouse1 contains a fact table named FactSales that has one billion rows.

You run the following T-SQL statement.

```
CREATE TABLE test.FactSales AS CLONE OF Dbo.FactSales;
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
A replica of <code>dbo.Sales</code> is created in the test schema by copying the metadata only.	☐	☐
Additional schema changes to <code>dbo.FactSales</code> will also apply to <code>test.FactSales</code> .	☐	☐
Additional schema changes to <code>dbo.FactSales</code> will also apply to <code>test.FactSales</code> .	☐	☐

Answer:

Answer Area

Statements	Yes	No
A replica of <code>dbo.Sales</code> is created in the test schema by copying the metadata only.	☒	☐
Additional schema changes to <code>dbo.FactSales</code> will also apply to <code>test.FactSales</code> .	☐	☒
Additional schema changes to <code>dbo.FactSales</code> will also apply to <code>test.FactSales</code> .	☐	☒

Explanation:

1. Yes

A replica of `dbo.Sales` is created in the test schema by copying the metadata only.

If metadata-only copying is used, it means the structure (schema) is copied but not the data.

This is possible with views, synonyms, or schema-only copies (e.g., `CREATE TABLE ... AS NO DATA`).

If the intent is to create an empty table with the same structure, this would be true.

2.No

Additional schema changes to `dbo.FactSales` will also apply to `test.FactSales`.

If `test.FactSales` was independently created (i.e., copied separately from `dbo.FactSales`), then any schema changes to `dbo.FactSales` will NOT automatically reflect in `test.FactSales`.

Schema changes would only propagate if `test.FactSales` was a synonym, view, or linked table.

Since it's likely an independent table, changes to `dbo.FactSales` do not apply automatically.

3.No

Additional schema changes to `dbo.FactSales` will also apply to `test.FactSales`.

This statement is identical to statement 2, so the answer remains the same.

Changes to `dbo.FactSales` do not automatically apply to `test.FactSales` unless explicitly linked (e.g., via a view or a synonym).

Question: 40

CertyIQ

You have source data in a folder on a local computer.

You need to create a solution that will use Fabric to populate a data store. The solution must meet the following requirements:

Support the use of dataflows to load and append data to the data store.

Ensure that Delta tables are V-Order optimized and compacted automatically.

Which type of data store should you use?

- A.a lakehouse
- B.an Azure SQL database
- C.a warehouse
- D.a KQL database

Answer: A

Explanation:

The Lakehouse and the Delta Lake table format are central to Microsoft Fabric, assuring that tables are optimized for analytics is a key requirement.

To meet the requirements of supporting dataflows to load and append data to the data store while ensuring that Delta tables are V-Order optimized and compacted automatically, you should use a lakehouse in Fabric as your solution.

Reference:

<https://learn.microsoft.com/en-us/fabric/data-engineering/delta-optimization-and-v-order?tabs=sparksql>

Question: 41

CertyIQ

HOTSPOT -
You have a Fabric tenant that contains a lakehouse.
You are using a Fabric notebook to save a large DataFrame by using the following code. df.write.partitionBy("year", "month", "day").mode("overwrite").parquet("Files/SalesOrder")
For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
The results will form a hierarchy of folders for each partition key.	☐	☐
The resulting file partitions can be read in parallel across multiple nodes.	☐	☐
The resulting file partitions will use file compression.	☐	☐

Answer:

Answer Area

Statements	Yes	No
The results will form a hierarchy of folders for each partition key.	☒	☐
The resulting file partitions can be read in parallel across multiple nodes.	☒	☐
The resulting file partitions will use file compression.	☒	☐

Explanation:

1. "The results will form a hierarchy of folders for each partition key."

Yes

Partitioning creates a folder structure where each folder represents a partition key value.

For example, in a data lake, partitioned data may be stored as:

bash

Copy

Edit

/sales/year=2023/month=01/

/sales/year=2023/month=02/

This improves query performance by allowing queries to skip unnecessary partitions.

2. "The resulting file partitions can be read in parallel across multiple nodes."

Yes

Partitioned files are designed for parallel processing.

In distributed environments like Spark, Synapse, or Fabric Lakehouse, each partition can be processed independently across multiple compute nodes, improving performance.

This is especially true for columnar formats like Parquet and Delta tables, which support distributed query execution.

3."The resulting file partitions will use file compression."

Yes

File compression is commonly used in partitioned storage to reduce storage costs and improve read performance.

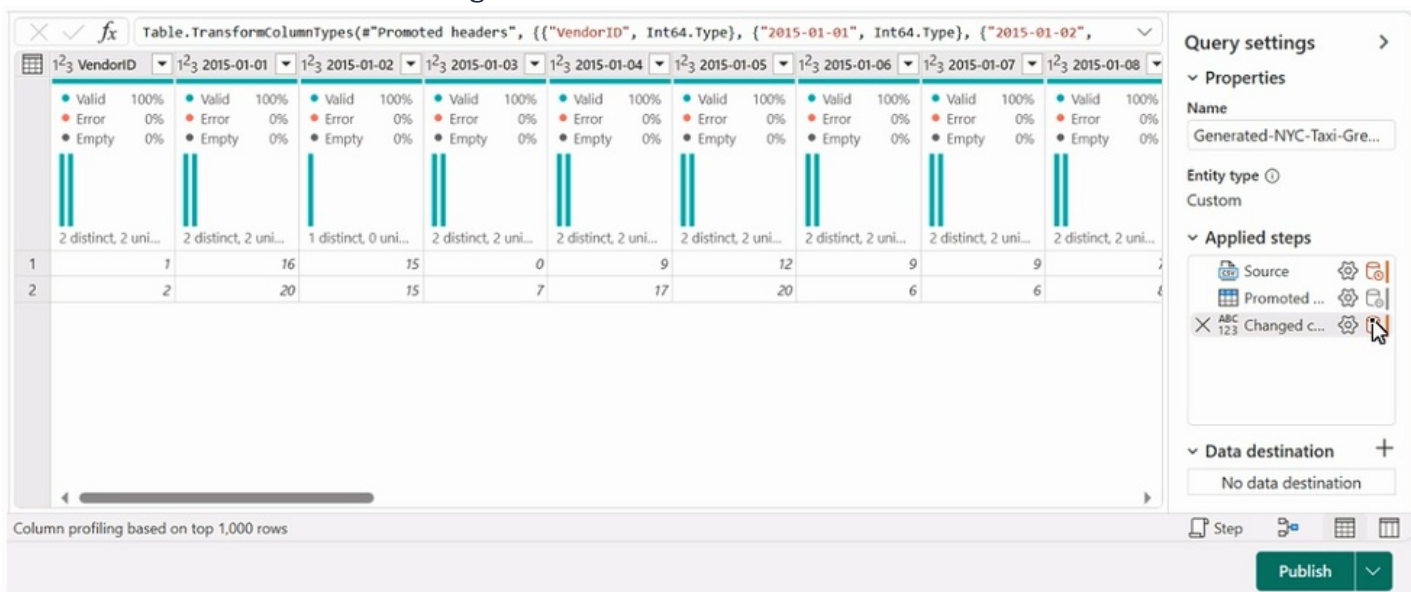
Formats like Parquet, ORC, and Delta support built-in compression (e.g., Snappy, Gzip, or ZSTD).

Compressed partitions reduce I/O costs and improve query performance, making them a best practice.

Question: 42

CertyIQ

You have a Fabric workspace named Workspace1 that contains a data flow named Dataflow1 contains a query that returns the data shown in the following exhibit.



You need to transform the data columns into attribute-value pairs, where columns become rows.

You select the VendorID column.

Which transformation should you select from the context menu of the VendorID column?

- A.Group by
- B.Unpivot columns
- C.Unpivot other columns
- D.Split column
- E.Remove other columns

Answer: C

Explanation:

C. Unpivot Other Columns.

You need to transform data columns into attribute-value pairs, meaning that columns will be converted into rows. This is achieved using Unpivoting in Power Query.

"Unpivot Columns" (Option B) → Converts selected columns into rows.

"Unpivot Other Columns" (Option C) → Converts all columns except the selected column into rows.

Since you are selecting the VendorID column and want to transform the other columns into attribute-value pairs, you should choose "Unpivot Other Columns."

Question: 43

CertyIQ

You have a Fabric tenant that contains a data pipeline.
You need to ensure that the pipeline runs every four hours on Mondays and Fridays.
To what should you set Repeat for the schedule?

- A.Daily
- B.By the minute
- C.Weekly
- D.Hourly

Answer: C

Explanation:

Answer **C : Weekly**.

The only way to do this is to set the schedule to "Weekly", set the days on Monday and Friday and add manually 6 Time of 4 hour intervals.

You need the pipeline to run every four hours on Mondays and Fridays.

Weekly scheduling allows you to specify particular days of the week (e.g., Monday and Friday).

Within the weekly schedule, you can set the pipeline to repeat every 4 hours.

Question: 44

CertyIQ

You have a Fabric tenant that contains a warehouse.
Several times a day, the performance of all warehouse queries degrades. You suspect that Fabric is throttling the compute used by the warehouse.
What should you use to identify whether throttling is occurring?

- A.the Capacity settings
- B.the Monitoring hub
- C.dynamic management views (DMVs)
- D.the Microsoft Fabric Capacity Metrics app

Answer: D

Explanation:

The Microsoft Capacity Metrics app, also known as the metrics app, serves as a monitoring tool within the Microsoft Power BI service. It offers functionalities to track and analyze the resource utilization.

Reference:

<https://learn.microsoft.com/en-us/fabric/data-warehouse/compute-capacity-smoothing-throttling>

Question: 45

CertyIQ

HOTSPOT -

You have a Fabric workspace that uses the default Spark starter pool and runtime version 1.2.

You plan to read a CSV file named Sales_raw.csv in a lakehouse, select columns, and save the data as a Delta table to the managed area of the lakehouse. Sales_raw.csv contains 12 columns.

You have the following code.

```
from pyspark.sql.functions import year

(spark
 .red
 .format("csv")
 .option("header", 'true')
 .load("Files/sales_raw.csv")
 .select('SalesOrderNumber', 'OrderDate', 'CustomerName',
 'UnitPrice')
 .withColumn("Year", year("OrderDate"))
 .write
 .partitionBy('Year')
 .saveAsTable("sales")
)
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
The Spark engine will read only the 'SalesOrderNumber', 'OrderDate', 'CustomerName', 'UnitPrice' columns from Sales_raw.csv.	☐	☐
Removing the partition will reduce the execution time of the query.	☐	☐
Adding inferSchema= 'true' to the options will increase the execution time of the query.	☐	☐

Answer:

Answer Area

Statements	Yes	No
The Spark engine will read only the 'SalesOrderNumber', 'OrderDate', 'CustomerName', 'UnitPrice' columns from Sales_raw.csv.	☐	☒
Removing the partition will reduce the execution time of the query.	☐	☒
Adding <code>inferSchema= 'true'</code> to the options will increase the execution time of the query.	☒	☐

Explanation:

No: The Spark engine will initially read all columns from the CSV file because the `.select()` transformation is applied after the data has been read into memory. Therefore, all 12 columns from Sales_raw.csv are read before the selection of specific columns is applied.

•No: Removing the partition might not necessarily reduce the execution time. While there might be some overhead in writing data to partitions, the overall impact on read performance, especially for large datasets, is usually beneficial. The query execution time for saving might be higher due to partitioning, but the read performance improvement usually outweighs this cost.

•Yes: Adding `inferSchema = 'true'` will increase the execution time of the query because Spark will need to read through the entire dataset to determine the data types of each column. This extra pass over the data adds to the initial read time.

Question: 46

CertyIQ

You have a Fabric tenant that contains a warehouse.

A user discovers that a report that usually takes two minutes to render has been running for 45 minutes and has still not rendered.

You need to identify what is preventing the report query from completing.

Which dynamic management view (DMV) should you use?

- A.sys.dm_exec_requests
- B.sys.dm_exec_sessions
- C.sys.dm_exec_connections
- D.sys.dm_pdw_exec_requests

Answer: A

Explanation:

The `sys.dm_exec_requests` dynamic management view provides real-time information about all requests currently executing on the SQL Server. This view helps you understand the status of each request, including whether it is blocked, waiting, or running.

Reference:

Question: 47

DRAG DROP -

You are creating a data flow in Fabric to ingest data from an Azure SQL database by using a T-SQL statement. You need to ensure that any foldable Power Query transformation steps are processed by the Microsoft SQL Server engine.

How should you complete the code? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Values

Answer Area

EnableFolding

NativeQuery

Optimize

Record

StopFolding

Table

Value

```
let
    Source = Sql.Databases(
        "server.database.windows.net"
    ),
    Database = Source{[Name = "db"]}[Data],
    Query = [ ] . [ ] (
        Database,
        " SELECT * FROM customer WHERE country IN ('USA', 'UK')",
        null,
        [ [ ] = true]
    )
in
    Query
```

Answer:

Values

Answer Area

EnableFolding

NativeQuery

Optimize

Record

StopFolding

Table

Value

```
let
    Source = Sql.Databases(
        "server.database.windows.net"
    ),
    Database = Source{[Name = "db"]}[Data],
    Query = Value . NativeQuery (
        Database,
        " SELECT * FROM customer WHERE country IN ('USA', 'UK')",
        null,
        [ EnableFolding = true]
    )
in
    Query
```

Explanation:

Value.

Value is the function used to store the result of the query.

NativeQuery.

NativeQuery allows executing a direct SQL query within Power Query M code.

EnableFolding .

This enables query folding, ensuring that transformations are executed in SQL Server rather than in Power Query.

Question: 48

CertyIQ

DRAG DROP -

You have a Fabric tenant that contains a lakehouse named Lakehouse1.

Readings from 100 IoT devices are appended to a Delta table in Lakehouse1. Each set of readings is approximately 25 KB. Approximately 10 GB of data is received daily.

All the table and SparkSession settings are set to the default.

You discover that queries are slow to execute. In addition, the lakehouse storage contains data and log files that are no longer used.

You need to remove the files that are no longer used and combine small files into larger files with a target size of 1 GB per file.

What should you do? To answer, drag the appropriate actions to the correct requirements. Each action may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Actions

Answer Area

Set the autoCompact
table setting.

Set the optimizeWrite
table setting.

Run the VACUUM
command on a schedule.

Set the autoCompact
SparkSession setting.

Run the OPTIMIZE
command on a schedule.

Set the parallelDelete
SparkSession setting.

Remove the files:

Combine the files:

Answer:

Answer Area

Remove the files:

Run the VACUUM command on a schedule.

Combine the files:

Run the OPTIMIZE command on a schedule.

Explanation:

Remove the files that are no longer used:

Run the VACUUM command on a schedule: The VACUUM command cleans up old files and log files that are no longer needed by the Delta table, helping to free up storage and potentially improve performance by reducing the number of files the query engine needs to consider.

Combine small files into larger files:

Run the OPTIMIZE command on a schedule: The OPTIMIZE command compacts small files into larger ones, improving read performance by reducing the overhead associated with opening many small files. This can be particularly useful when you have a large number of small files due to frequent appends of small data sets.

Question: 49

CertyIQ

You need to create a data loading pattern for a Type 1 slowly changing dimension (SCD). Which two actions should you include in the process? Each correct answer presents part of the solution. NOTE: Each correct answer is worth one point.

- A. Update rows when the non-key attributes have changed.
- B. Insert new rows when the natural key exists in the dimension table, and the non-key attribute values have changed.
- C. Update the effective end date of rows when the non-key attribute values have changed.
- D. Insert new records when the natural key is a new value in the table.

Answer: AD

Explanation:

A. Update rows when the non-key attributes have changed: In a Type 1 SCD, when a change is detected in any of the non-key attributes of an existing row, the current row is updated with the new values. This type of SCD

does not keep any historical data; it simply overwrites the old data with the new data.

D. Insert new records when the natural key is a new value in the table: When a new record (with a new natural key) is encountered that does not already exist in the dimension table, it is inserted as a new row. This is necessary to ensure that all new entities are captured in the dimension.

Question: 50

CertyIQ

HOTSPOT -

You have a Fabric workspace named Workspace1 and an Azure Data Lake Storage Gen2 account named storage1. Workspace1 contains a lakehouse named Lakehouse1.

You need to create a shortcut to storage1 in Lakehouse1.

Which connection and endpoint should you specify? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Connection:

	▼
abfs	
abfss	
https	

Endpoint:

	▼
blob	
dfs	
file	

Answer:

Answer Area

Connection:

	▼
abfs	
abfss	
https	

Endpoint:

	▼
blob	
dfs	
file	

Explanation:

1. Connection Type: **https**

The selection of https means that the connection is being made using secure HTTPS endpoints rather than using Azure Blob File System (ABFS/ABFSS) protocols.

https is commonly used for REST API-based access to Azure Storage.

2. Endpoint Type: **dfs**

The selection of dfs (Distributed File System) means that the connection is using Azure Data Lake Storage Gen2.

dfs endpoints enable Hierarchical Namespace (HNS), which allows for file system-like directory structures and optimized big data analytics.

This is required for Spark, Azure Synapse, Fabric, and Databricks.

Question: 51

CertyIQ

You are analyzing customer purchases in a Fabric notebook by using PySpark.

You have the following DataFrames:

transactions: Contains five columns named transaction_id, customer_id, product_id, amount, and date and has 10 million rows, with each row representing a transaction. customers: Contains customer details in 1,000 rows and three columns named customer_id, name, and country.

You need to join the DataFrames on the customer_id column. The solution must minimize data shuffling.

You write the following code.

```
from pyspark.sql import functions as F
```

```
results =
```

Which code should you run to populate the results DataFrame?

- A.transactions.join(F.broadcast(customers), transactions.customer_id == customers.customer_id)
- B.transactions.join(customers, transactions.customer_id == customers.customer_id).distinct()
- C.transactions.join(customers, transactions.customer_id == customers.customer_id)
- D.transactions.crossJoin(customers).where(transactions.customer_id == customers.customer_id)

Answer: A

Explanation:

A. transactions.join(F.broadcast(customers), transactions.customer_id == customers.customer_id) Optimized method to perform a join between a very large table and a smaller one.

In Apache Spark, broadcasting refers to an optimization technique for join operations. When you join two DataFrames or RDDs and one of them is significantly smaller than the other, Spark can "broadcast" the smaller table to all nodes in the cluster. This approach avoids the need for network shuffles for each row of the larger table, significantly reducing the execution time of the join operation.

<https://sparkbyexamples.com/spark/broadcast-join-in-spark/>

Question: 52

CertyIQ

HOTSPOT -

You have a Microsoft Power BI report and a semantic model that uses Direct Lake mode. From Power BI Desktop, you open Performance analyzer as shown in the following exhibit.

Performance analyzer

Start recording Refresh visuals Stop

Clear Export

Name	Duration (ms)
Recording started (5/15/2023 5:10:49 PM)	-
Refreshed visual	-
Card	638
DAX query	431
Visual display	25
Other	181
Copy query	
Changed a filter	-
Edited a visual	-
Card	46
Changed a filter	-
Card	9205
DAX query	9055
Direct query	8660
Visual display	24
Other	126
Copy query	

Visualizations

Build visual

Fields

Sum of col1

Drill through

Cross-report

Keep all filters

Add drill-through fields here

Use the drop-down menus to select the answer choice that completes each statement based on the information presented in the graphic.
NOTE: Each correct selection is worth one point.

Answer Area

The Direct Lake fallback behavior is set to

Automatic
DirectLakeOnly
DirectQueryOnly

The query for the table visual is executed by using

the composite model
Direct Lake
Direct Query

Answer:

Answer Area

The Direct Lake fallback behavior is set to

▼

Automatic

DirectLakeOnly

DirectQueryOnly

The query for the table visual is executed by using

▼

the composite model

Direct Lake

Direct Query

Explanation:

1. Direct Lake Fallback Behavior: **Automatic.**

Direct Lake Mode enables real-time querying of Delta tables in a Fabric lakehouse without requiring data import or DirectQuery.

When fallback behavior is set to Automatic, Power BI will automatically switch between Direct Lake Mode and DirectQuery when necessary (e.g., if a feature isn't supported in Direct Lake).

This provides the best balance between performance and flexibility.

2. Query Execution: **Direct Lake.**

Direct Lake Mode allows Power BI to load data directly from Delta tables in a Fabric lakehouse without preloading or sending queries to an external database.

It provides better performance than DirectQuery by eliminating query overhead and improving response times.

Question: 53

CertyIQ

HOTSPOT -

You have a Fabric tenant that contains a lakehouse named Lakehouse1. Lakehouse1 contains a table named Nyctaxi_raw. Nyctaxi_raw contains the following table:

Name	Data type
pickupDateTime	Timestamp
passengerCount	Integer
fareAmount	Double
paymentType	String
tipAmount	Double

You create a Fabric notebook and attach it to Lakehouse1.

You need to use PySpark code to transform the data. The solution must meet the following requirements:

Add a column named pickupDate that will contain only the date portion of pickupDateTime.

Filter the DataFrame to include only rows where fareAmount is a positive number that is less than 100.

How should you complete the code? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
df = spark.read.format("delta").load("Tables/nyctaxi_raw")
```

```
df2=  ("pickupDate", df['tpepPickupDateTime'])
```

- df.columns
- df.select
- df.withColumn
- df.withColumnsRenamed

- .alias('date')
- .cast('date')
- .cast('pickupDate')
- .getfield('date')

- .filter("fareAmount > 0 AND farAmount < 100")
- .filter(col("fareAmount").contains("1..100"))
- .when(df.fareAmount > 0 AND fareAmount < 100)
- .where(df.fareAmount.isin([1,100]))

Answer:

Answer Area

```
df = spark.read.format("delta").load("Tables/nyctaxi_raw")
```

```
df2=  ("pickupDate", df['tpepPickupDateTime'])
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- df.columns
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- df.withColumn
- df.withColumnsRenamed

- .alias('date')
- .cast('date')
- .cast('pickupDate')
- .getfield('date')

- .filter("fareAmount > 0 AND farAmount < 100")
- .filter(col("fareAmount").contains("1..100"))
- .when(df.fareAmount > 0 AND fareAmount < 100)
- .where(df.fareAmount.isin([1,100]))

Explanation:**1. df.withColumn.**

The withColumn function is used to add or replace a column in a DataFrame.

It is appropriate here because we are creating a new column (pickupDate) based on an existing column (tpepPickupDateTime).

Alternative options like df.select or df.withColumnsRenamed would not modify a column directly.

2 .cast('date')

The .cast('date') function is used to convert a column's data type to date.

Since tpepPickupDateTime is likely a timestamp, converting it to a date ensures it only contains the date portion.

Other options like .alias('date') only rename the column without changing its type.

3.filter("fareAmount > 0 AND fareAmount < 100")

The .filter() function in PySpark is used to apply SQL-like filtering conditions to DataFrames.

The selected condition ensures that only rows where fareAmount is greater than 0 and less than 100 are retained.

Other options like .where(df.fareAmount.isin([1,100])) filter for specific values rather than a range.

Question: 54**CertyIQ**

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a Fabric tenant that contains a new semantic model in OneLake.

You use a Fabric notebook to read the data into a Spark DataFrame.

You need to evaluate the data to calculate the min, max, mean, and standard deviation values for all the string and numeric columns.

Solution: You use the following PySpark expression:

```
df.explain()
```

Does this meet the goal?

A.Yes

B.No

Answer: B**Explanation:**

The df.explain() method in PySpark is used to print the logical and physical plans of a DataFrame, which helps in understanding how Spark plans to execute the query. It does not compute any statistical values like min, max, mean, or standard deviation.

To achieve the goal, you should use: df.describe().show()

Question: 55

CertyIQ

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

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You use a Fabric notebook to read the data into a Spark DataFrame.

You need to evaluate the data to calculate the min, max, mean, and standard deviation values for all the string and numeric columns.

Solution: You use the following PySpark expression:

```
df.show()
```

Does this meet the goal?

A.Yes

B.No

Answer: B

Explanation:

Correct methods: Use `df.describe().show()` for basic statistics and `df.agg()` with appropriate functions (min, max, mean, stddev) for detailed statistics.

The correct syntax is `df.describe()`. Sources: * describe -->

<https://spark.apache.org/docs/latest/api/python/reference/pyspark.sql/api/pyspark.sql.DataFrame.describe.html>*
show -->

<https://spark.apache.org/docs/latest/api/python/reference/pyspark.sql/api/pyspark.sql.DataFrame.show.html>

Question: 56

CertyIQ

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

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You have a Fabric tenant that contains a new semantic model in OneLake.

You use a Fabric notebook to read the data into a Spark DataFrame.

You need to evaluate the data to calculate the min, max, mean, and standard deviation values for all the string and numeric columns.

Solution: You use the following PySpark expression:

```
df.summary()
```

Does this meet the goal?

A.Yes

B.No

Answer: A

Explanation:

A. Yes

The `df.summary()` function in PySpark provides summary statistics for both numeric and string columns in a `DataFrame`, including:

For numeric columns: count, mean, stddev, min, max

For string columns: count, min, max

Question: 57

CertyIQ

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a Fabric tenant that contains a lakehouse named Lakehouse1. Lakehouse1 contains a Delta table named Customer.

When you query Customer, you discover that the query is slow to execute. You suspect that maintenance was NOT performed on the table.

You need to identify whether maintenance tasks were performed on Customer.

Solution: You run the following Spark SQL statement:

`DESCRIBE HISTORY customer -`

Does this meet the goal?

A.Yes

B.No

Answer: A

Explanation:

Yes, running `DESCRIBE HISTORY customer` meets the goal of identifying whether maintenance tasks were performed on the Delta table.

Question: 58

CertyIQ

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a Fabric tenant that contains a lakehouse named Lakehouse1. Lakehouse1 contains a Delta table named Customer.

When you query Customer, you discover that the query is slow to execute. You suspect that maintenance was NOT performed on the table.

You need to identify whether maintenance tasks were performed on Customer.

Solution: You run the following Spark SQL statement:

`REFRESH TABLE customer -`

Does this meet the goal?

A.Yes

B.No

Answer: B

Explanation:

No, running REFRESH TABLE customer does not meet the goal of identifying whether maintenance tasks were performed on the Delta table. This Spark SQL command is used to refresh the metadata of a table. It ensures that the latest schema and data are available for queries but does not give any historical information about maintenance operations.

Question: 59

CertyIQ

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

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When you query Customer, you discover that the query is slow to execute. You suspect that maintenance was NOT performed on the table.

You need to identify whether maintenance tasks were performed on Customer.

Solution: You run the following Spark SQL statement:

EXPLAIN TABLE customer -

Does this meet the goal?

A.Yes

B.No

Answer: B

Explanation:

The EXPLAIN statement in Spark SQL is used to display the execution plan of a query. This plan shows how Spark will execute the query, including details about the operations and stages involved. While it is useful for understanding and optimizing query performance, it does not provide historical information about maintenance tasks like optimization, compaction, or vacuuming performed on the table.

Question: 60

CertyIQ

Case study -

This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam. You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided.

To answer the questions included in a case study, you will need to reference information that is provided in the case study. Case studies might contain exhibits and other resources that provide more information about the scenario that is described in the case study. Each question is independent of the other questions in this case study. At the end of this case study, a review screen will appear. This screen allows you to review your answers and to make changes before you move to the next section of the exam. After you begin a new section, you cannot return to this section.

To start the case study -

To display the first question in this case study, click the Next button. Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment, and problem statements. If the case study has an All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs. When you are ready to answer a question, click the Question button to return to the question.

Overview -

Litware, Inc. is a manufacturing company that has offices throughout North America. The analytics team at Litware contains data engineers, analytics engineers, data analysts, and data scientists.

Existing Environment -

Fabric Environment -

Litware has been using a Microsoft Power BI tenant for three years. Litware has NOT enabled any Fabric capacities and features.

Available Data -

Litware has data that must be analyzed as shown in the following table.

Description	Original source	Total size
Customer data	Customer relationship management (CRM) system	50 MB
Product data	Customer relationship management (CRM) system	200 MB
Customer satisfaction surveys	SurveyMonkey	500 GB

The Product data contains a single table and the following columns.

Name	Data type
ProductID	Integer
ProductName	String
ProductCategory	String
ListPrice	Decimal

The customer satisfaction data contains the following tables:

- Survey
- Question
- Response

For each survey submitted, the following occurs:

- One row is added to the Survey table.
- One row is added to the Response table for each question in the survey.

The Question table contains the text of each survey question. The third question in each survey response is an overall satisfaction score. Customers can submit a survey after each purchase.

User Problems -

The analytics team has large volumes of data, some of which is semi-structured. The team wants to use Fabric to create a new data store.

Product data is often classified into three pricing groups: high, medium, and low. This logic is implemented in several databases and semantic models, but the logic does NOT always match across implementations.

Requirements -

Planned Changes -

Litware plans to enable Fabric features in the existing tenant. The analytics team will create a new data store as a proof of concept (PoC). The remaining Litware users will only get access to the Fabric features once the PoC is complete. The PoC will be completed by using a Fabric trial capacity

The following three workspaces will be created:

- AnalyticsPOC: Will contain the data store, semantic models, reports pipelines, dataflow, and notebooks used to populate the data store
- DataEngPOC: Will contain all the pipelines, dataflows, and notebooks used to populate OneLake
- DataSciPOC: Will contain all the notebooks and reports created by the data scientists

The following will be created in the AnalyticsPOC workspace:

- A data store (type to be decided)
- A custom semantic model
- A default semantic model
- Interactive reports

The data engineers will create data pipelines to load data to OneLake either hourly or daily depending on the data source. The analytics engineers will create processes to ingest, transform, and load the data to the data store in the AnalyticsPOC workspace daily. Whenever possible, the data engineers will use low-code tools for data ingestion. The choice of which data cleansing and transformation tools to use will be at the data engineers' discretion.

All the semantic models and reports in the Analytics POC workspace will use the data store as the sole data source.

Technical Requirements -

The data store must support the following:

- Read access by using T-SQL or Python
- Semi-structured and unstructured data
- Row-level security (RLS) for users executing T-SQL queries

Files loaded by the data engineers to OneLake will be stored in the Parquet format and will meet Delta Lake specifications.

Data will be loaded without transformation in one area of the AnalyticsPOC data store. The data will then be cleansed, merged, and transformed into a dimensional model

The data load process must ensure that the raw and cleansed data is updated completely before populating the dimensional model

The dimensional model must contain a date dimension. There is no existing data source for the date dimension. The Litware fiscal year matches the calendar year. The date dimension must always contain dates from 2010 through the end of the current year.

The product pricing group logic must be maintained by the analytics engineers in a single location. The pricing group data must be made available in the data store for T-SQL queries and in the default semantic model. The following logic must be used:

- List prices that are less than or equal to 50 are in the low pricing group.
- List prices that are greater than 50 and less than or equal to 1,000 are in the medium pricing group.
- List prices that are greater than 1,000 are in the high pricing group.

Security Requirements -

Only Fabric administrators and the analytics team must be able to see the Fabric items created as part of the PoC. Litware identifies the following security requirements for the Fabric items in the AnalyticsPOC workspace:

- Fabric administrators will be the workspace administrators.
- The data engineers must be able to read from and write to the data store. No access must be granted to datasets or reports.
- The analytics engineers must be able to read from, write to, and create schemas in the data store. They also must be able to create and share semantic models with the data analysts and view and modify all reports in the workspace.
- The data scientists must be able to read from the data store, but not write to it. They will access the data by using a Spark notebook
- The data analysts must have read access to only the dimensional model objects in the data store. They also must have access to create Power BI reports by using the semantic models created by the analytics engineers.
- The date dimension must be available to all users of the data store.
- The principle of least privilege must be followed.

Both the default and custom semantic models must include only tables or views from the dimensional model in the data store. Litware already has the following Microsoft Entra security groups:

- FabricAdmins: Fabric administrators
- AnalyticsTeam: All the members of the analytics team
- DataAnalysts: The data analysts on the analytics team
- DataScientists: The data scientists on the analytics team
- DataEngineers: The data engineers on the analytics team
- AnalyticsEngineers: The analytics engineers on the analytics team

Report Requirements -

The data analysts must create a customer satisfaction report that meets the following requirements:

- Enables a user to select a product to filter customer survey responses to only those who have purchased that product.
- Displays the average overall satisfaction score of all the surveys submitted during the last 12 months up to a selected date.
- Shows data as soon as the data is updated in the data store.
- Ensures that the report and the semantic model only contain data from the current and previous year.
- Ensures that the report respects any table-level security specified in the source data store.
- Minimizes the execution time of report queries.

You need to recommend a solution to prepare the tenant for the PoC.

Which two actions should you recommend performing from the Fabric Admin portal? Each correct answer presents part of the solution.

NOTE: Each correct answer is worth one point.

- A.Enable the Users can try Microsoft Fabric paid features option for the entire organization.
- B.Enable the Users can try Microsoft Fabric paid features option for specific security groups.
- C.Enable the Allow Azure Active Directory guest users to access Microsoft Fabric option for specific security groups.
- D.Enable the Users can create Fabric items option and exclude specific security groups.
- E.Enable the Users can create Fabric items option for specific security groups.

Answer: BE

Explanation:

Enable the Users can try Microsoft Fabric paid features option for specific security groups: This will allow specific security groups (like the AnalyticsTeam, DataAnalysts, DataScientists, DataEngineers, and AnalyticsEngineers) to access and use the paid features of Microsoft Fabric necessary for the PoC. This is important to ensure that only the relevant team members can utilize these advanced features while preventing unnecessary access for other users.

Enable the Users can create Fabric items option for specific security groups: This will allow only specific security groups to create Fabric items, ensuring that the creation of these items is controlled and managed by the appropriate team members. This helps maintain the principle of least privilege and ensures that only authorized personnel can create and manage Fabric items during the PoC.

Question: 61

CertyIQ

HOTSPOT

-
Case study

-
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-
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All the semantic models and reports in the Analytics POC workspace will use the data store as the sole data source.

Technical Requirements

-
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- The analytics engineers must be able to read from, write to, and create schemas in the data store. They also must be able to create and share semantic models with the data analysts and view and modify all reports in the workspace.
- The data scientists must be able to read from the data store, but not write to it. They will access the data by using a Spark notebook
- The data analysts must have read access to only the dimensional model objects in the data store. They also must have access to create Power BI reports by using the semantic models created by the analytics engineers.
- The date dimension must be available to all users of the data store.
- The principle of least privilege must be followed.

Both the default and custom semantic models must include only tables or views from the dimensional model in the data store. Litware already has the following Microsoft Entra security groups:

- FabricAdmins: Fabric administrators
- AnalyticsTeam: All the members of the analytics team
- DataAnalysts: The data analysts on the analytics team
- DataScientists: The data scientists on the analytics team
- DataEngineers: The data engineers on the analytics team
- AnalyticsEngineers: The analytics engineers on the analytics team

Report Requirements

-

The data analysts must create a customer satisfaction report that meets the following requirements:

- Enables a user to select a product to filter customer survey responses to only those who have purchased that product.
- Displays the average overall satisfaction score of all the surveys submitted during the last 12 months up to a selected date.
- Shows data as soon as the data is updated in the data store.
- Ensures that the report and the semantic model only contain data from the current and previous year.
- Ensures that the report respects any table-level security specified in the source data store.
- Minimizes the execution time of report queries.

You need to design a semantic model for the customer satisfaction report.

Which data source authentication method and mode should you use? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Authentication method:

Basic authentication
Service principal authentication
Single sign-on (SSO) authentication

Mode:

Direct Lake
DirectQuery
Import

Answer:

Answer Area

Authentication method:

Basic authentication
Service principal authentication
Single sign-on (SSO) authentication

Mode:

Direct Lake
DirectQuery
Import

Explanation:

1. Authentication Method: "**Single sign-on (SSO) authentication**"

SSO (Single Sign-On) allows users to authenticate once and gain access to multiple applications without needing to log in separately.

This is a secure and seamless authentication method commonly used in enterprise environments for Microsoft Fabric, Power BI, and other services.

Compared to Basic authentication (which requires username/password) or Service principal authentication (used for automated access), SSO provides better user experience and security.

2. Mode: "**Direct Lake**"

Direct Lake mode is a new hybrid query mode that combines the performance of Import mode with the real-

time access of DirectQuery.

Unlike Import mode, which loads data into memory, or DirectQuery, which queries the source live, Direct Lake queries Delta tables directly from OneLake without the need for pre-loading.

This mode is optimized for high performance and scalability, making it an ideal choice for Power BI and Fabric scenarios.

Question: 62

CertyIQ

Case study -

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Overview -

Litware, Inc. is a manufacturing company that has offices throughout North America. The analytics team at Litware contains data engineers, analytics engineers, data analysts, and data scientists.

Existing Environment -

Fabric Environment -

Litware has been using a Microsoft Power BI tenant for three years. Litware has NOT enabled any Fabric capacities and features.

Available Data -

Litware has data that must be analyzed as shown in the following table.

Description	Original source	Total size
Customer data	Customer relationship management (CRM) system	50 MB
Product data	Customer relationship management (CRM) system	200 MB
Customer satisfaction surveys	SurveyMonkey	500 GB

The Product data contains a single table and the following columns.

Name	Data type
ProductID	Integer
ProductName	String
ProductCategory	String
ListPrice	Decimal

The customer satisfaction data contains the following tables:

- Survey
- Question
- Response

For each survey submitted, the following occurs:

- One row is added to the Survey table.
- One row is added to the Response table for each question in the survey.

The Question table contains the text of each survey question. The third question in each survey response is an overall satisfaction score. Customers can submit a survey after each purchase.

User Problems -

The analytics team has large volumes of data, some of which is semi-structured. The team wants to use Fabric to create a new data store.

Product data is often classified into three pricing groups: high, medium, and low. This logic is implemented in several databases and semantic models, but the logic does NOT always match across implementations.

Requirements -

Planned Changes -

Litware plans to enable Fabric features in the existing tenant. The analytics team will create a new data store as a proof of concept (PoC). The remaining Liware users will only get access to the Fabric features once the PoC is complete. The PoC will be completed by using a Fabric trial capacity

The following three workspaces will be created:

- AnalyticsPOC: Will contain the data store, semantic models, reports pipelines, dataflow, and notebooks used to populate the data store

- DataEngPOC: Will contain all the pipelines, dataflows, and notebooks used to populate OneLake

- DataSciPOC: Will contain all the notebooks and reports created by the data scientists

The following will be created in the AnalyticsPOC workspace:

- A data store (type to be decided)

- A custom semantic model

- A default semantic model

- Interactive reports

The data engineers will create data pipelines to load data to OneLake either hourly or daily depending on the data source. The analytics engineers will create processes to ingest, transform, and load the data to the data store in the AnalyticsPOC workspace daily. Whenever possible, the data engineers will use low-code tools for data ingestion. The choice of which data cleansing and transformation tools to use will be at the data engineers' discretion.

All the semantic models and reports in the Analytics POC workspace will use the data store as the sole data source.

Technical Requirements -

The data store must support the following:

- Read access by using T-SQL or Python

- Semi-structured and unstructured data

- Row-level security (RLS) for users executing T-SQL queries

Files loaded by the data engineers to OneLake will be stored in the Parquet format and will meet Delta Lake specifications.

Data will be loaded without transformation in one area of the AnalyticsPOC data store. The data will then be cleansed, merged, and transformed into a dimensional model

The data load process must ensure that the raw and cleansed data is updated completely before populating the dimensional model

The dimensional model must contain a date dimension. There is no existing data source for the date dimension. The Litware fiscal year matches the calendar year. The date dimension must always contain dates from 2010 through the end of the current year.

The product pricing group logic must be maintained by the analytics engineers in a single location. The pricing group data must be made available in the data store for T-SOL. queries and in the default semantic model. The following logic must be used:

- List prices that are less than or equal to 50 are in the low pricing group.

- List prices that are greater than 50 and less than or equal to 1,000 are in the medium pricing group.

- List prices that are greater than 1,000 are in the high pricing group.

Security Requirements -

Only Fabric administrators and the analytics team must be able to see the Fabric items created as part of the PoC.

Litware identifies the following security requirements for the Fabric items in the AnalyticsPOC workspace:

- Fabric administrators will be the workspace administrators.

- The data engineers must be able to read from and write to the data store. No access must be granted to datasets or reports.

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- Shows data as soon as the data is updated in the data store.
- Ensures that the report and the semantic model only contain data from the current and previous year.
- Ensures that the report respects any table-level security specified in the source data store.
- Minimizes the execution time of report queries.

You need to implement the date dimension in the data store. The solution must meet the technical requirements.

What are two ways to achieve the goal? Each correct answer presents a complete solution.

NOTE: Each correct selection is worth one point.

- A.Populate the date dimension table by using a dataflow.
- B.Populate the date dimension table by using a Copy activity in a pipeline.
- C.Populate the date dimension view by using T-SQL.
- D.Populate the date dimension table by using a Stored procedure activity in a pipeline.

Answer: AD

Explanation:

A. Populate the date dimension table by using a dataflow.

Dataflows in Microsoft Fabric (or Power BI) allow ETL (Extract, Transform, Load) operations to create and populate tables.

They can be used to generate and maintain a date dimension with calculated columns such as year, quarter, month, and week.

D. Populate the date dimension table by using a Stored procedure activity in a pipeline.

Stored procedures allow for predefined logic and transformations to be applied while inserting or updating records in the date dimension table.

Using a Stored Procedure activity in an Azure Data Factory (ADF) or Microsoft Fabric pipeline is a common way to populate and maintain a date dimension.

Question: 63

CertyIQ

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Report Requirements -

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- Shows data as soon as the data is updated in the data store.
- Ensures that the report and the semantic model only contain data from the current and previous year.
- Ensures that the report respects any table-level security specified in the source data store.
- Minimizes the execution time of report queries.

You need to ensure the data loading activities in the AnalyticsPOC workspace are executed in the appropriate sequence. The solution must meet the technical requirements.

What should you do?

- A.Create a dataflow that has multiple steps and schedule the dataflow.
- B.Create and schedule a Spark notebook.
- C.Create and schedule a Spark job definition.

D.Create a pipeline that has dependencies between activities and schedule the pipeline.

Answer: D

Explanation:

D. Create a pipeline that has dependencies between activities and schedule the pipeline.

To ensure that data loading activities in the AnalyticsPOC workspace are executed in the appropriate sequence while meeting technical requirements, the best approach is to use a data pipeline with dependencies between activities.

Pipelines in Microsoft Fabric (or Azure Data Factory) allow you to orchestrate data movement, transformation, and processing across multiple steps.

You can define activity dependencies, ensuring that each step executes only after the previous step completes successfully.

You can schedule the pipeline to run at specific times, making it automated and reliable.

Question: 64

CertyIQ

Case study -

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Overview -

Contoso, Ltd. is a US-based health supplements company. Contoso has two divisions named Sales and Research. The Sales division contains two departments named Online Sales and Retail Sales. The Research division assigns internally developed product lines to individual teams of researchers and analysts.

Existing Environment -

Identity Environment -

Contoso has a Microsoft Entra tenant named contoso.com. The tenant contains two groups named ResearchReviewersGroup1 and ResearchReviewersGroup2.

Data Environment -

Contoso has the following data environment:

- The Sales division uses a Microsoft Power BI Premium capacity.
- The semantic model of the Online Sales department includes a fact table named Orders that uses Import mode. In the system of origin, the OrderID value represents the sequence in which orders are created.
- The Research department uses an on-premises, third-party data warehousing product.
- Fabric is enabled for contoso.com.
- An Azure Data Lake Storage Gen2 storage account named storage1 contains Research division data for a product line named Productline1. The data is in the delta format.
- A Data Lake Storage Gen2 storage account named storage2 contains Research division data for a product line named Productline2. The data is in the CSV format.

Requirements -

Planned Changes -

Contoso plans to make the following changes:

- Enable support for Fabric in the Power BI Premium capacity used by the Sales division.
- Make all the data for the Sales division and the Research division available in Fabric.
- For the Research division, create two Fabric workspaces named Productline1ws and Productine2ws.
- In Productline1ws, create a lakehouse named Lakehouse1.
- In Lakehouse1, create a shortcut to storage1 named ResearchProduct.

Data Analytics Requirements -

Contoso identifies the following data analytics requirements:

- All the workspaces for the Sales division and the Research division must support all Fabric experiences.
- The Research division workspaces must use a dedicated, on-demand capacity that has per-minute billing.
- The Research division workspaces must be grouped together logically to support OneLake data hub filtering based on the department name.
- For the Research division workspaces, the members of ResearchReviewersGroup1 must be able to read lakehouse and warehouse data and shortcuts by using SQL endpoints.
- For the Research division workspaces, the members of ResearchReviewersGroup2 must be able to read lakehouse data by using Lakehouse explorer.
- All the semantic models and reports for the Research division must use version control that supports branching.

Data Preparation Requirements -

Contoso identifies the following data preparation requirements:

- The Research division data for Productline1 must be retrieved from Lakehouse1 by using Fabric notebooks.
- All the Research division data in the lakehouses must be presented as managed tables in Lakehouse explorer.

Semantic Model Requirements -

Contoso identifies the following requirements for implementing and managing semantic models:

- The number of rows added to the Orders table during refreshes must be minimized.
- The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -

Contoso identifies the following high-level requirements that must be considered for all solutions:

- Follow the principle of least privilege when applicable.
- Minimize implementation and maintenance effort when possible.

You need to recommend which type of Fabric capacity SKU meets the data analytics requirements for the Research division.

What should you recommend?

- A.A
- B.EM
- C.P
- D.F

Answer: D

Explanation:

Fabric capacity SKUs all start with F, they allows for Pay as you go by the minute. Reserved capacity is also available.

Reference:

<https://azure.microsoft.com/en-us/pricing/details/microsoft-fabric/>

Question: 65

CertyIQ

HOTSPOT

-

Case study

-

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Requirements

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Planned Changes

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General Requirements

Contoso identifies the following high-level requirements that must be considered for all solutions:

- Follow the principle of least privilege when applicable.
- Minimize implementation and maintenance effort when possible.

You need to migrate the Research division data for Productline1. The solution must meet the data preparation requirements.

How should you complete the code? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
df=spark.read.format('csv').options(header="true",inferSchema="true").load("fs1@storage1.dfs.core.windows.net/files/productline1")  
df.write.mode("overwrite").format(" ").save(" ")
```

csv
delta
parquet

productline1
Tables/productline1
Tables/research/productline1

Answer:

Answer Area

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csv
delta
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productline1
Tables/productline1
tables/research/productline1

Explanation:

- "delta"

- "Tables/productline1"

Requirements: Use managed tables.

If you use `saveAsTable()` you don't need to specify the path "Table/"

If you use `save()` you specify the full path

Question: 66

CertyIQ

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- A Data Lake Storage Gen2 storage account named storage2 contains Research division data for a product line named Productline2. The data is in the CSV format.

Requirements -

Planned Changes -

Contoso plans to make the following changes:

- Enable support for Fabric in the Power BI Premium capacity used by the Sales division.
- Make all the data for the Sales division and the Research division available in Fabric.
- For the Research division, create two Fabric workspaces named Productline1ws and Productline2ws.
- In Productline1ws, create a lakehouse named Lakehouse1.
- In Lakehouse1, create a shortcut to storage1 named ResearchProduct.

Data Analytics Requirements -

Contoso identifies the following data analytics requirements:

- All the workspaces for the Sales division and the Research division must support all Fabric experiences.
- The Research division workspaces must use a dedicated, on-demand capacity that has per-minute billing.
- The Research division workspaces must be grouped together logically to support OneLake data hub filtering based on the department name.
- For the Research division workspaces, the members of ResearchReviewersGroup1 must be able to read lakehouse and warehouse data and shortcuts by using SQL endpoints.
- For the Research division workspaces, the members of ResearchReviewersGroup2 must be able to read lakehouse data by using Lakehouse explorer.
- All the semantic models and reports for the Research division must use version control that supports branching.

Data Preparation Requirements -

Contoso identifies the following data preparation requirements:

- The Research division data for Productline1 must be retrieved from Lakehouse1 by using Fabric notebooks.
- All the Research division data in the lakehouses must be presented as managed tables in Lakehouse explorer.

Semantic Model Requirements -

Contoso identifies the following requirements for implementing and managing semantic models:

- The number of rows added to the Orders table during refreshes must be minimized.
- The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements -

Contoso identifies the following high-level requirements that must be considered for all solutions:

- Follow the principle of least privilege when applicable.
- Minimize implementation and maintenance effort when possible.

What should you use to implement calculation groups for the Research division semantic models?

- A. Microsoft Power BI Desktop
- B. the Power BI service
- C. DAX Studio
- D. Tabular Editor

Answer: D

Explanation:

D. Tabular Editor.

The semantic models in the Research division workspaces must use Direct Lake mode.

Considering Contoso's requirement to minimize implementation and maintenance effort (general requirement), Tabular Editor offers a more efficient way to define calculation groups compared to manually writing DAX code. Additionally, since calculation groups are part of the semantic model itself, they can be deployed and managed alongside the model, simplifying maintenance.

Question: 67

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HOTSPOT

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Case study

-
This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam. You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided. To answer the questions included in a case study, you will need to reference information that is provided in the case study. Case studies might contain exhibits and other resources that provide more information about the scenario that is described in the case study. Each question is independent of the other questions in this case study. At the end of this case study, a review screen will appear. This screen allows you to review your answers and to make changes before you move to the next section of the exam. After you begin a new section, you cannot return to this section.

To start the case study

-
To display the first question in this case study, click the Next button. Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment, and problem statements. If the case study has an All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs. When you are ready to answer a question, click the Question button to return to the question.

Overview

-
Contoso, Ltd. is a US-based health supplements company. Contoso has two divisions named Sales and Research. The Sales division contains two departments named Online Sales and Retail Sales. The Research division assigns internally developed product lines to individual teams of researchers and analysts.

Existing Environment

-
Identity Environment

-
Contoso has a Microsoft Entra tenant named contoso.com. The tenant contains two groups named ResearchReviewersGroup1 and ResearchReviewersGroup2.

Data Environment

-
Contoso has the following data environment:

- The Sales division uses a Microsoft Power BI Premium capacity.
- The semantic model of the Online Sales department includes a fact table named Orders that uses Import mode. In the system of origin, the OrderID value represents the sequence in which orders are created.
- The Research department uses an on-premises, third-party data warehousing product.
- Fabric is enabled for contoso.com.
- An Azure Data Lake Storage Gen2 storage account named storage1 contains Research division data for a product line named Productline1. The data is in the delta format.
- A Data Lake Storage Gen2 storage account named storage2 contains Research division data for a product line named Productline2. The data is in the CSV format.

Requirements

-
Planned Changes

-
Contoso plans to make the following changes:

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Semantic Model Requirements

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The number of rows added to the Orders table during refreshes must be minimized.

- The semantic models in the Research division workspaces must use Direct Lake mode.

General Requirements

-

Contoso identifies the following high-level requirements that must be considered for all solutions:

- Follow the principle of least privilege when applicable.
- Minimize implementation and maintenance effort when possible.

Which workspace role assignments should you recommend for ResearchReviewersGroup1 and ResearchReviewersGroup2? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

ResearchReviewersGroup1:

Contributor
Member
Viewer

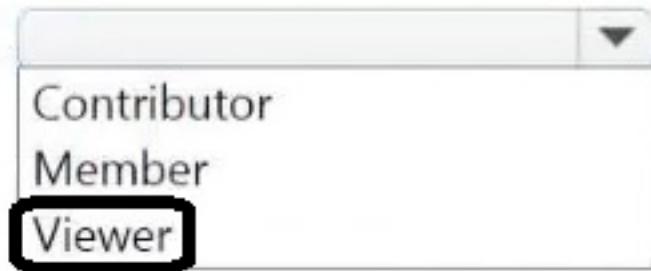
ResearchReviewersGroup2:

Contributor
Member
Viewer

Answer:

Answer Area

ResearchReviewersGroup1:



A dropdown menu for ResearchReviewersGroup1. The menu is open, showing three options: Contributor, Member, and Viewer. The Viewer option is highlighted with a black rectangular box.

ResearchReviewersGroup2:



A dropdown menu for ResearchReviewersGroup2. The menu is open, showing three options: Contributor, Member, and Viewer. The Contributor option is highlighted with a black rectangular box.

Explanation:

1. ResearchReviewersGroup1: **Viewer**.

Members of this group can only view resources but cannot edit, add, or delete them.

Suitable for users who need read-only access to reports, datasets, or workspaces.

2. ResearchReviewersGroup2: **Contributor**.

Members can edit and add content but cannot manage access or settings.

Suitable for users who need to modify or update reports and datasets but should not have full administrative privileges.